List R2 — Valuation: free cash flows, discount rates, and the NPV rule

Worked solutions

Statements: `Listas/lista-r2-valuation.md` (copied verbatim from the books); each exercise's full text is reproduced right before its solution. Each section cites its exercise by book, number and *printed* page of the local editions: Berk, DeMarzo & Harford, *Fundamentals of Corporate Finance*, 2nd ed. (2011); Koski, Higgins & Mitton, *Analysis for Financial Management*, 13th ed. (2023); Grinblatt & Titman, *Financial Markets and Corporate Strategy*, 2nd ed. (2002); Ross, Westerfield, Jordan & Lamb, *Fundamentos de Administração Financeira*, 9th ed. (2013); Amaro de Matos, *Theoretical Foundations of Corporate Finance* (2001). Rates are written as decimals inside equations. Each exercise follows the same order: what is asked, approach, notation, steps with a boxed answer per part, intuition, common mistakes, book answer and reference. Every number below is reproduced by `Listas/lista-r2-valuation-solutions_check.py`.

1. NPV of a two-year investment — Berk 8.2, p. 241

Statement.

Source: Berk–DeMarzo–Harford, Ch. 8, Exercise 2, p. 241 (PDF 272) · Answer in book: no

You have an opportunity to invest \$100,000 now in return for \$80,000 in one year and \$30,000 in two years. If your cost of capital is 9%, what is the NPV of this investment?

What is asked. Find the value created today by paying \$100,000 now to receive \$80,000 in one year and \$30,000 in two years, when money costs 9% a year, and say whether to invest.

Approach. The NPV rule: discount every cash flow to $t = 0$ at the cost of capital and add them. The cost of capital is the return investors give up by not investing elsewhere at the same risk, so a positive NPV means the project beats that alternative.

Notation.

- $C_0 = 100{,}000$: investment paid today, in dollars ($t = 0$).
- $C_1 = 80{,}000$, $C_2 = 30{,}000$: cash inflows at $t = 1$ and $t = 2$, in dollars.
- $r = 0.09$: cost of capital per year, $r > 0$.
- NPV: net present value at $t = 0$, in dollars, any real number.

Step 1 — NPV definition. Each future dollar is divided by $(1 + r)^t$ to bring it to today; the outflow is already at $t = 0$ and enters with a minus sign:

$$\text{NPV} = -C_0 + \frac{C_1}{1 + r} + \frac{C_2}{(1 + r)^2}. \tag{1.1}$$

Step 2 — Discount factors. The two-year factor compounds the rate twice:

$$(1.09)^2 = (1.09)(1.09) = 1.1881. \tag{1.2}$$

Step 3 — Substitute the data. Placing the numbers and (1.2) in (1.1):

$$\begin{aligned} \text{NPV} &= -(100{,}000) + \frac{(80{,}000)}{(1.09)} + \frac{(30{,}000)}{(1.1881)} \\ &= -100{,}000 + 73{,}394.50 + 25{,}250.40 \\ &= -100{,}000 + 98{,}644.90 \\ &= -1{,}355.10. \end{aligned} \tag{1.3}$$

Step 4 — Decision. By (1.3) the NPV is negative: the inflows are worth \$98,644.90 today, less than the \$100,000 they cost. Reject the investment.

Answer (NPV). NPV $= -\$1{,}355.10 < 0$: do not invest.

Intuition. The inflows sum to \$110,000, a 10% total return over two years, while 9% per year compounded would require $1.1881 - 1 = 18.81\%$ over two years. Money returned later is worth less, so the project destroys \$1,355.10 of value relative to investing at the cost of capital.

Common mistakes.

- Adding undiscounted cash flows ($110{,}000 - 100{,}000 = +10{,}000$) and accepting.
- Discounting the year-2 flow by $1.09 \times 2 = 2.18$ or by 1.18 instead of $(1.09)^2$.
- Discounting C_0 as well: it is paid today, its discount factor is 1.

Ref: Berk, §8.1–8.2 (The NPV decision rule; using the NPV rule), pp. 212–216.

2. CAPM: Intel and Boeing — Berk 12.24, p. 377

Statement.

Source: Berk–DeMarzo–Harford, Ch. 12, Exercise 24, p. 377 (PDF 408) · Answer in book: no

Suppose Intel stock has a beta of 1.6, whereas Boeing stock has a beta of 1. If the risk-free interest rate is 4% and the expected return of the market portfolio is 10%, according to the CAPM,

a. What is the expected return of Intel stock?

b. What is the expected return of Boeing stock?

c. What is the beta of a portfolio that consists of 60% Intel stock and 40% Boeing stock?

d. What is the expected return of a portfolio that consists of 60% Intel stock and 40% Boeing stock (show both ways to solve this)?

What is asked. (a) and (b): the expected return the CAPM assigns to Intel (beta 1.6) and Boeing (beta 1). (c): the beta of a 60/40 Intel–Boeing portfolio. (d): that portfolio's expected return, computed once from its beta and once as the average of the two stocks' returns.

Approach. The security market line prices every asset by its beta: expected return equals the risk-free rate plus beta times the market risk premium. Beta and expected return are both linear in portfolio weights, which is why (d) has two routes that must agree.

Notation.

- $r_f = 0.04$: risk-free rate; $E[R_m] = 0.10$: expected market return; $E[R_m] - r_f$: market risk premium.
- $\beta_I = 1.6$, $\beta_B = 1.0$: betas of Intel and Boeing stock, real numbers.
- $E[R_I]$, $E[R_B]$, $E[R_P]$: expected returns of Intel, Boeing and the portfolio P.
- $w_I = 0.6$, $w_B = 0.4$: portfolio weights, $w_I + w_B = 1$; β_P: beta of P.

Step 1 — The CAPM (security market line). For any security i, the compensation for risk is its beta times the premium of the market over the risk-free rate:

$$E[R_i] = r_f + \beta_i \, (E[R_m] - r_f). \tag{2.1}$$

The market risk premium is

$$E[R_m] - r_f = 0.10 - 0.04 = 0.06. \tag{2.2}$$

Step 2 — (a) Intel. Substituting β_I and (2.2) into (2.1):

$$\begin{aligned} E[R_I] &= (0.04) + (1.6)(0.06) \\ &= 0.04 + 0.096 \\ &= 0.136 = 13.6\%. \end{aligned} \tag{2.3}$$

Answer (a). $E[R_I] = 13.6\%$.

Step 3 — (b) Boeing. Substituting β_B and (2.2) into (2.1):

$$\begin{aligned} E[R_B] &= (0.04) + (1.0)(0.06) \\ &= 0.04 + 0.06 \\ &= 0.10 = 10\%. \end{aligned} \tag{2.4}$$

Answer (b). $E[R_B] = 10\%$ (beta 1: the market's return).

Step 4 — (c) Portfolio beta. The portfolio's return is $w_I R_I + w_B R_B$, and covariance with the market is linear, so beta is the weighted average of the betas:

$$\beta_P = w_I \beta_I + w_B \beta_B \tag{2.5}$$

$$\begin{aligned} &= (0.6)(1.6) + (0.4)(1.0) \\ &= 0.96 + 0.40 \\ &= 1.36. \end{aligned} \tag{2.6}$$

Answer (c). $\beta_P = 1.36$.

Step 5 — (d) Portfolio expected return, two ways. First, substituting β_P from (2.6) and (2.2) into (2.1):

$$\begin{aligned} E[R_P] &= (0.04) + (1.36)(0.06) \\ &= 0.04 + 0.0816 \\ &= 0.1216 = 12.16\%. \end{aligned} \tag{2.7}$$

Second, as the weighted average of (2.3) and (2.4):

$$\begin{aligned} E[R_P] &= w_I E[R_I] + w_B E[R_B] \\ &= (0.6)(0.136) + (0.4)(0.10) \\ &= 0.0816 + 0.04 \\ &= 0.1216 = 12.16\%. \end{aligned} \tag{2.8}$$

The two routes agree because both (2.1) and (2.5) are linear in the weights: $w_I(r_f + \beta_I \cdot 0.06) + w_B(r_f + \beta_B \cdot 0.06) = (w_I + w_B)r_f + (w_I\beta_I + w_B\beta_B)\,0.06$, and $w_I + w_B = 1$.

Answer (d). $E[R_P] = 12.16\%$ by both routes.

Intuition. Under the CAPM only market risk is priced, and it is measured by beta. Intel moves 1.6 times as much as the market, so it earns 1.6 risk premiums. A portfolio is priced by its average beta, so averaging returns or averaging betas gives the same answer.

Common mistakes.

- Multiplying beta by the market *return* (10%) instead of the *premium* (6%): $0.04 + 1.6(0.10) = 20\%$ is wrong.
- Averaging the betas with equal weights (1.3) instead of the 60/40 weights.
- Forgetting r_f in route one of (d), or adding it twice when averaging returns.

Ref: Berk, §12.1 (expected return of a portfolio), p. 347; §12.4 (the CAPM), pp. 366–372.

3. Coffee shop: the initial cash flow — Berk 9.18, p. 276

Statement.

Source: Berk–DeMarzo–Harford, Ch. 9, Exercise 18, p. 276 (PDF 307) · Answer in book: no

You have just completed a $20,000 feasibility study for a new coffee shop in some retail space you own. You bought the space two years ago for $100,000, but if you sold it today, you would net $115,000 after taxes. Outfitting the space for a coffee shop would require a capital expenditure of $30,000 plus an initial investment of $5,000 in inventory. What is the correct initial cash flow for your analysis of the coffee shop opportunity?

What is asked. Find the single cash flow at $t = 0$ that belongs in the coffee-shop NPV, deciding which of the five dollar amounts given (study, purchase price, resale value, capex, inventory) count.

Approach. The incremental-cash-flow principle: include a cash flow only if it changes because the project is taken. Sunk costs are out (they happen either way); opportunity costs are in (using an asset you own means not selling it); working capital is an investment like capex.

Notation.

- $S = 20{,}000$: feasibility study already paid (sunk). $P = 100{,}000$: price paid for the space two years ago (sunk).
- $L = 115{,}000$: after-tax amount the space would fetch if sold today (opportunity cost).
- $K = 30{,}000$: capital expenditure on outfitting; $N = 5{,}000$: inventory (investment in net working capital).
- CF_0: incremental cash flow at $t = 0$, in dollars.

Step 1 — Incremental principle. Compare the two worlds, "open the shop" and "do not". S and P have already been paid in both, so they cancel. Opening the shop means giving up the sale of the space, so the after-tax sale value L is lost only in the "open" world, as are K and N:

$$\mathrm{CF}_0 = -(L + K + N). \tag{3.1}$$

Step 2 — Substitute. Placing the numbers in (3.1):

$$\begin{aligned} \mathrm{CF}_0 &= -\big((115{,}000) + (30{,}000) + (5{,}000)\big) \\ &= -150{,}000. \end{aligned} \tag{3.2}$$

Answer (CF_0). $\mathrm{CF}_0 = -\$150{,}000$.

Intuition. The question is "what do I give up today to run the shop?". The answer is the $115,000 I could pocket by selling the space, plus the new money for equipment and stock. The study and the old purchase price are gone either way.

Common mistakes.

- Including the $20,000 study ($-170{,}000$): it is sunk.
- Using the historical $100,000 instead of today's after-tax $115,000 as the cost of the space, or leaving the space out because "we already own it".
- Forgetting the $5,000 inventory: working capital ties up cash just like equipment.

Ref: Berk, §9.3–9.4 (incremental free cash flow; opportunity costs, sunk costs), pp. 254–265.

4. Pre-tax WACC and the return on assets — Berk 13.4, p. 403

Statement.

Source: Berk–DeMarzo–Harford, Ch. 13, Exercise 4, p. 403 (PDF 434) · Answer in book: no

Consider a simple firm that has the following market-value balance sheet:

Assets		Liabilities and Equity	
	1000	Debt	400
		Equity	600

Next year, there are two possible values for its assets, each equally likely: \$1200 and \$960. Its debt will be due with 5% interest. Because all of the cash flows from the assets must go to either the debt or the equity, if you hold a portfolio of the debt and equity in the same proportions as the firm's capital structure, your portfolio should earn exactly the expected return on the firm's assets. Show that a portfolio invested 40% in the firm's debt and 60% in its equity will have the same expected return as the assets of the firm. That is, show that the firm's pre-tax WACC is the same as the expected return on its assets.

What is asked. Show, with the numbers of the balance sheet, that a portfolio holding 40% debt and 60% equity of this firm earns the same expected return as the firm's assets, i.e. that the pre-tax WACC equals the expected return on assets.

Approach. Compute each claim's payoff in each state (debt first, equity is the residual), turn payoffs into expected returns, and compare the weighted average with the assets' return. Then prove it in general: the claims add up to the assets state by state (value additivity), so their value-weighted return is the assets' return.

Notation.

- $A_0 = 1000$, $D_0 = 400$, $E_0 = 600$: market values today of assets, debt and equity, $A_0 = D_0 + E_0$.
- $A_1^H = 1200$, $A_1^L = 960$: asset values next year, each with probability 1/2.
- $r_D = 0.05$: promised interest rate on debt; D_1: payment to debt next year.
- E_1^H, E_1^L: equity payoffs next year; r_E, r_A: expected returns on equity and assets.
- $w_D = D_0/A_0 = 0.4$, $w_E = E_0/A_0 = 0.6$: capital-structure weights.

Step 1 — Debt payoff. Debt holders are promised principal plus 5% interest, $D_0(1 + r_D)$:

$$\begin{aligned} D_1 &= (400)(1.05) \\ &= 420. \end{aligned} \tag{4.1}$$

Since $A_1^L = 960 > 420$, the promise is paid in both states, so debt is riskless and its expected return is $r_D = 0.05$.

Step 2 — Equity payoffs. Equity receives what is left after debt, $E_1^s = A_1^s - D_1$. Substituting (4.1):

$$\begin{aligned} E_1^H &= 1200 - (420) = 780, \\ E_1^L &= 960 - (420) = 540. \end{aligned} \tag{4.2}$$

Step 3 — Expected equity return. Average the payoffs (4.2) with probability 1/2 each, then

divide by today's price:

$$\begin{aligned} E[E_1] &= (0.5)(780) + (0.5)(540) \\ &= 390 + 270 \\ &= 660, \\ r_E &= \frac{(660)}{(600)} - 1 \\ &= 1.10 - 1 \\ &= 0.10. \end{aligned} \tag{4.3}$$

Step 4 — Expected asset return. The same two operations for the assets:

$$\begin{aligned} E[A_1] &= (0.5)(1200) + (0.5)(960) \\ &= 600 + 480 \\ &= 1080, \\ r_A &= \frac{(1080)}{(1000)} - 1 \\ &= 1.08 - 1 \\ &= 0.08. \end{aligned} \tag{4.4}$$

Step 5 — Return on the 40/60 portfolio (pre-tax WACC). A portfolio's return is the weighted average of its holdings' returns. Substituting r_D and (4.3):

$$\begin{aligned} w_D r_D + w_E r_E &= (0.4)(0.05) + (0.6)(0.10) \\ &= 0.02 + 0.06 \\ &= 0.08, \end{aligned} \tag{4.5}$$

which equals r_A in (4.4).

Step 6 — Why it must hold (general proof). Write each return as expected payoff over price minus one, with $w_D = D_0/A_0$ and $w_E = E_0/A_0$, and multiply out:

$$\begin{aligned} w_D r_D + w_E r_E &= \frac{D_0}{A_0}\Big(\frac{E[D_1]}{D_0} - 1\Big) + \frac{E_0}{A_0}\Big(\frac{E[E_1]}{E_0} - 1\Big) \\ &= \frac{E[D_1]}{A_0} - \frac{D_0}{A_0} + \frac{E[E_1]}{A_0} - \frac{E_0}{A_0} \\ &= \frac{E[D_1] + E[E_1]}{A_0} - \frac{D_0 + E_0}{A_0} \\ &= \frac{E[A_1]}{A_0} - 1 = r_A, \end{aligned} \tag{4.6}$$

where the last line uses $D_1 + E_1 = A_1$ state by state (all asset cash flows go to debt or equity, so $E[D_1] + E[E_1] = E[A_1]$) and $D_0 + E_0 = A_0$.

Answer (proof). $r_D = 5\%$, $r_E = 10\%$, $0.4(5\%) + 0.6(10\%) = 8\% = r_A$.

Intuition. Holding 40% of the debt and 60% of the equity is holding 100% of the claims on the assets, scaled down. A portfolio that owns the whole cash flow of the assets must earn what the assets earn; leverage only splits that return between the two claims.

Common mistakes.

- Using book weights or the payoffs next year as weights; the weights are today's market values, 400/1000 and 600/1000.

- Computing equity's return on the payoff difference $(780 - 540)$ or forgetting to subtract the *interest* as well as the principal $(1200 - 400)$.
- Applying the after-tax WACC: there are no taxes here, and the identity is about the *pre-tax* WACC.

Ref: Berk, §13.1 (a first look at the WACC), pp. 383–386; §13.3, p. 390.

5. WACC for SKYE Corporation — Higgins 8.7, p. 328

Statement.

Source: Higgins (Koski–Higgins–Mitton), Ch. 8, Exercise 7, p. 328 (PDF 349) · Answer in book: yes, p. 422

What is the weighted-average cost of capital for SKYE Corporation given the following information?

Equity shares outstanding	1 million
Stock price per share	\$30.00
Yield to maturity on debt	7.68%
Book value of interest-bearing debt	\$10 million
Coupon interest rate on debt	9%
Interest rate on government bonds	6%
SKYE's equity beta	0.75
Historical excess return on stocks	6.3%
Tax rate	40%

What is asked. Compute SKYE's weighted-average cost of capital from the table: decide which debt rate (coupon or yield), which weights (book or market) and which equity cost to use.

Approach. Higgins' Equation 8.1, $K_W = [(1-t)K_D D + K_E E]/(D+E)$: each source of capital weighted by its value, debt at its after-tax cost because interest is tax deductible. The cost of equity comes from the CAPM (Higgins' Eq. 8.2); the cost of debt is the yield, the return a buyer of the debt earns today.

Notation.

- E: market value of equity (millions of dollars); D: value of interest-bearing debt (millions), $D, E > 0$.
- $K_D = 0.0768$: pre-tax cost of debt (yield to maturity); $t = 0.40$: tax rate, $0 \le t < 1$.
- $r_f = 0.06$: government bond rate; $\beta = 0.75$: equity beta; MRP $= 0.063$: historical excess return on stocks.
- K_E: cost of equity; K_W: weighted-average cost of capital.

Step 1 — Weights. Equity at market value is shares times price:

$$E = (1)(30) = 30, \qquad D = 10, \qquad D + E = 10 + 30 = 40. \tag{5.1}$$

The market value of the debt is not given (no maturity), so, as Higgins does, use its book value $D = 10$.

Step 2 — Cost of equity (CAPM). Risk-free rate plus beta times the excess return on stocks:

$$\begin{aligned} K_E &= r_f + \beta\,\text{MRP} && (5.2)\\ &= (0.06) + (0.75)(0.063)\\ &= 0.06 + 0.04725\\ &= 0.10725. && (5.3) \end{aligned}$$

Step 3 — After-tax cost of debt. The relevant rate is the yield, not the 9% coupon; each dollar of interest saves t dollars of tax, so the firm's net cost is $(1-t)K_D$:

$$\begin{aligned}(1-t)K_D &= (1-0.40)(0.0768)\\ &= (0.60)(0.0768)\\ &= 0.04608. \end{aligned} \tag{5.4}$$

Step 4 — WACC. Higgins' Equation 8.1, $K_W = \dfrac{(1-t)K_D D + K_E E}{D+E}$. Substituting (5.4), (5.3) and (5.1):

$$\begin{aligned}K_W &= \frac{(0.04608)(10)+(0.10725)(30)}{(40)}\\ &= \frac{0.4608+3.2175}{40}\\ &= \frac{3.6783}{40}\\ &= 0.0919575 \approx 9.2\%. \end{aligned} \tag{5.5}$$

Answer (WACC). $K_W \approx 9.2\%$.

Intuition. The firm's capital is three-quarters equity, so the WACC sits close to the 10.7% cost of equity, pulled down by cheap debt whose cost is cut further by the interest tax deduction.

Remark. The 9% coupon above the 7.68% yield means the bonds trade above par, so the market value of debt exceeds \$10 million and the true market-weight WACC is slightly lower. Without a maturity the market value cannot be computed; book value is the stated fallback.

Common mistakes.

- Using the 9% coupon as the cost of debt: the coupon was set at issue; the yield is today's required return.
- Forgetting the tax adjustment $(1-t)$ on debt, or applying it to equity too.
- Weighting equity at book value; the price times shares (\$30 million) is the market value.

Book answer. Matches the book's answer, $K_W = 9.2\%$ (p. 422).

Ref: Higgins, §8.4.1–8.4.2 (cost of capital defined; cost of capital for Polaris, CAPM), pp. 295–304.

6. GM–Hughes: cost of capital from comparables — Grinblatt 11.10–11.13, p. 416

Statement.

Source: Grinblatt–Titman, Ch. 11, Exercises 11.10–11.13, p. 416 (PDF 429) · Answer in book: no

In 1989, General Motors (GM) was evaluating the acquisition of Hughes Aircraft Corporation. Recognizing that the appropriate discount rate for the projected cash flows of Hughes was different than its own cost of capital, GM assumed that Hughes had approximately the same risk as Lockheed or Northrop, which had low-risk defense contracts and products that were similar to Hughes. Specifically, assume the following inputs:

Comparison	β_E	D/E
GM	1.20	.40
Lockheed	0.90	.90
Northrop	0.85	.70

Target $\frac{D}{E}$ for Hughes's acquisition = 1

Hughes's expected cash flow next year = \$300 million
Growth rate of Hughes's cash flows = 5 percent per year
Appropriate discount rate on debt (riskless rate) = 8 percent
Expected return of the tangency portfolio = 14 percent

11.10. Analyze the Hughes acquisition (which took place) by first computing the betas of the comparison firms, Lockheed and Northrop, as if they were all equity financed. Assume no taxes.

11.11. Compute the beta of the assets of the Hughes acquisition, assuming no taxes, by taking the average of the asset betas of Lockheed and Northrop.

11.12. Compute the cost of capital for the Hughes acquisition, assuming no taxes.

11.13. Compute the value of Hughes with the cost of capital estimated in exercise 11.12.

What is asked. 11.10: strip the leverage out of Lockheed's and Northrop's equity betas to get the beta each would have if all-equity financed. 11.11: average those two asset betas to estimate the beta of Hughes' assets. 11.12: turn that beta into a discount rate. 11.13: value Hughes' growing cash flows at that rate.

Approach. The comparison (pure-play) method: the risk of a project is read off firms in the same business, not off the buyer. Their equity betas mix business risk with leverage, so we unlever: the asset beta is the value-weighted average of the debt and equity betas. Without taxes, MM says the asset cost of capital does not depend on financing, so the target D/E does not enter. The value then follows from the growing-perpetuity formula.

Notation.

- β_E: equity beta; D/E: debt-to-equity ratio (market values); β_A: asset (unlevered) beta.
- $\beta_D = 0$: debt beta (debt is riskless, discounted at r_f). No taxes.
- $r_f = 0.08$: riskless rate; $\bar{R}_T = 0.14$: expected return of the tangency portfolio.
- r_A: cost of capital of Hughes' assets; $C_1 = 300$: expected cash flow next year (\$ millions); $g = 0.05$: growth rate, $g < r_A$.
- V_H: value of Hughes (\$ millions).

Step 1 — Asset beta of a levered firm (11.10). Assets are a portfolio of the firm's debt and equity, so their beta is the value-weighted average:

$$\beta_A = \frac{E}{D+E}\,\beta_E + \frac{D}{D+E}\,\beta_D. \tag{6.1}$$

Substituting $\beta_D = 0$ removes the second term; dividing numerator and denominator by E:

$$\beta_A = \frac{E\,\beta_E}{D+E} = \frac{\beta_E}{(D+E)/E} = \frac{\beta_E}{1+D/E}. \tag{6.2}$$

Lockheed, substituting $\beta_E = 0.90$ and $D/E = 0.90$ into (6.2):

$$\beta_{A,L} = \frac{(0.90)}{1+(0.90)} = \frac{0.90}{1.90} = 0.4736842. \tag{6.3}$$

Northrop, substituting $\beta_E = 0.85$ and $D/E = 0.70$ into (6.2):

$$\beta_{A,N} = \frac{(0.85)}{1+(0.70)} = \frac{0.85}{1.70} = 0.5000000. \tag{6.4}$$

Answer (11.10). $\beta_{A,L} = 0.474$, $\beta_{A,N} = 0.500$.

Step 2 — Hughes asset beta (11.11). Averaging (6.3) and (6.4):

$$\begin{aligned}\beta_{A,H} &= \frac{(0.4736842)+(0.5000000)}{2}\\ &= \frac{0.9736842}{2}\\ &= 0.4868421.\end{aligned} \tag{6.5}$$

Answer (11.11). $\beta_{A,H} = 0.487$.

Step 3 — Cost of capital (11.12). With the CAPM ($\bar{R}_T$ plays the role of the market return), the premium is $\bar{R}_T - r_f = 0.14 - 0.08 = 0.06$. Substituting (6.5):

$$\begin{aligned} r_A &= r_f + \beta_{A,H}(\bar{R}_T - r_f) \qquad (6.6)\\ &= (0.08) + (0.4868421)(0.06) \\ &= 0.08 + 0.0292105 \\ &= 0.1092105 \approx 10.92\%. \qquad (6.7)\end{aligned}$$

The target $D/E = 1$ does not change this: without taxes the cost of the assets is independent of how they are financed (MM). Check: relevering (6.2) at $D/E = 1$ gives $\beta_{E,H} = (0.4868421)(1+1) = 0.9736842$, so $r_E = (0.08) + (0.9736842)(0.06) = 0.08 + 0.0584211 = 0.1384211$ and, with weights $D/V = E/V = 0.5$,

$$\begin{aligned} \text{WACC} &= (0.5)(0.08) + (0.5)(0.1384211) \\ &= 0.04 + 0.0692105 \\ &= 0.1092105, \end{aligned}$$

the same as (6.7).

Answer (11.12). $r_A \approx 10.92\%$.

Step 4 — Value of Hughes (11.13). The cash flow starts at C_1 next year and grows at g forever, so it is a growing perpetuity, valued at $C_1/(r-g)$ (requires $g < r$). Substituting (6.7):

$$\begin{aligned} V_H &= \frac{C_1}{r_A - g} \qquad (6.8)\\ &= \frac{(300)}{(0.1092105) - (0.05)} \\ &= \frac{300}{0.0592105} \\ &= 5{,}066.67 \text{ (\$ millions)}. \qquad (6.9)\end{aligned}$$

Answer (11.13). $V_H \approx$ \$5,066.67 million (≈ \$5.07 billion).

Remark. Using GM's own beta would be wrong: $\beta_{A,GM} = 1.20/1.40 = 0.8571429$, giving $r = (0.08) + (0.8571429)(0.06) = 0.08 + 0.0514286 = 0.1314286$ and $V = 300/(0.1314286 - 0.05) = 300/0.0814286 = 3{,}684.21$, undervaluing Hughes by about \$1.4 billion.

Intuition. The discount rate belongs to the project, not to the buyer. Hughes' risk is read off firms in the same business, after stripping out their leverage, because leverage scales equity risk without changing the risk of the operations.

Common mistakes.

- Averaging the *equity* betas (0.875) and using that as the asset beta: it keeps the comparables' leverage.
- Dividing by D/E instead of $1 + D/E$, or using D/V where the table gives D/E.
- Discounting with GM's cost of capital, or using C_0 instead of next year's C_1 in the growing perpetuity.

Ref: Grinblatt & Titman, §11.3 (the effect of leverage on comparisons), p. 379; §11.4 and Example 11.2 (comparison method, adjusting for leverage without taxes), pp. 384–386.

7. Mutually exclusive projects and the crossover rate — Berk 8.26, p. 244

Statement.

Source: Berk–DeMarzo–Harford, Ch. 8, Exercise 26, p. 244 (PDF 275) · Answer in book: no
You are considering the following two projects and can only take one. Your cost of capital is 11%.

	0	1	2	3	4
A	−100	25	30	40	50
B	−100	50	40	30	20

a. What is the NPV of each project at your cost of capital?

b. What is the IRR of each project?

c. At what cost of capital are you indifferent between the two projects?

d. What should you do?

What is asked. (a) The NPV of each project at 11%. (b) The discount rate that sets each project's NPV to zero. (c) The discount rate at which both projects have the same NPV. (d) Which one to take.

Approach. NPV at the cost of capital ranks mutually exclusive projects; IRR does not, because it ignores scale and timing. The crossover rate is found by subtracting one project from the other and computing the IRR of the difference: at that rate the incremental project is worth zero.

Notation.

- C_t^A, C_t^B: cash flows of projects A and B at $t = 0, \dots, 4$ (as tabulated in the statement).
- r: discount rate, $r > -1$; $r_0 = 0.11$: the cost of capital.
- $\mathrm{NPV}_j(r) = \sum_{t=0}^{4} C_t^j/(1+r)^t$, $j \in \{A, B\}$; IRR_j: root of $\mathrm{NPV}_j(r) = 0$.
- $\Delta_t = C_t^A - C_t^B$: incremental cash flows of A over B; r^*: crossover rate.

Step 1 — Discount factors at 11%. Each later factor multiplies the previous one by 1.11:

$$(1.11)^2 = 1.2321, \qquad (1.11)^3 = 1.367631, \qquad (1.11)^4 = 1.51807041. \tag{7.1}$$

Step 2 — (a) NPVs at 11%. Substituting r_0 and (7.1) in the definition:

$$\begin{aligned} \mathrm{NPV}_A &= -100 + \frac{25}{(1.11)} + \frac{30}{(1.2321)} + \frac{40}{(1.367631)} + \frac{50}{(1.51807041)} \\ &= -100 + 22.5225 + 24.3487 + 29.2477 + 32.9365 \\ &= 9.055, \end{aligned} \tag{7.2}$$

$$\begin{aligned} \mathrm{NPV}_B &= -100 + \frac{50}{(1.11)} + \frac{40}{(1.2321)} + \frac{30}{(1.367631)} + \frac{20}{(1.51807041)} \\ &= -100 + 45.0450 + 32.4649 + 21.9357 + 13.1746 \\ &= 12.620. \end{aligned} \tag{7.3}$$

Answer (a). $\mathrm{NPV}_A = 9.055$, $\mathrm{NPV}_B = 12.620$.

Step 3 — (b) IRRs. Both streams have one sign change (− then +), so $\mathrm{NPV}_j(r)$ is strictly decreasing in r and has exactly one root. Bracketing first, by evaluating the NPV formula at nearby rates:

$$\mathrm{NPV}_A(0.14) = 1.617 > 0, \qquad \mathrm{NPV}_A(0.15) = -0.688 < 0,$$
$$\mathrm{NPV}_B(0.17) = 1.360 > 0, \qquad \mathrm{NPV}_B(0.18) = -0.325 < 0,$$

so $\text{IRR}_A \in (14\%, 15\%)$ and $\text{IRR}_B \in (17\%, 18\%)$. Narrowing each interval (bisection, or a spreadsheet's IRR function):

$$\text{IRR}_A = 14.70\%, \qquad \text{IRR}_B = 17.80\%. \tag{7.4}$$

Answer (b). $\text{IRR}_A = 14.70\%$, $\text{IRR}_B = 17.80\%$.

Step 4 — (c) Crossover rate. Indifference means $\text{NPV}_A(r) = \text{NPV}_B(r)$, i.e. $\text{NPV}_A(r) - \text{NPV}_B(r) = 0$; since NPV is linear in the cash flows, this is the NPV of the incremental project Δ:

$$\Delta = (-100 - (-100),\ 25 - 50,\ 30 - 40,\ 40 - 30,\ 50 - 20) = (0,\ -25,\ -10,\ 10,\ 30),$$
$$0 = \frac{-25}{1+r} + \frac{-10}{(1+r)^2} + \frac{10}{(1+r)^3} + \frac{30}{(1+r)^4}. \tag{7.5}$$

Multiplying (7.5) by $(1+r)^4$ and writing $x = 1 + r > 0$:

$$-25x^3 - 10x^2 + 10x + 30 = 0. \tag{7.6}$$

The coefficients change sign once, so by Descartes' rule (7.6) has exactly one positive root. Bracketing with (7.5): $\text{NPV}_\Delta(0.05) = 0.440 > 0$ and $\text{NPV}_\Delta(0.06) = -0.326 < 0$, so $r^* \in (5\%, 6\%)$. Narrowing,

$$r^* = 5.57\%, \qquad \text{NPV}_A(r^*) = \text{NPV}_B(r^*) = 24.86. \tag{7.7}$$

For $r < r^*$, A has the higher NPV (at $r = 0$: $\text{NPV}_A = 45$ against $\text{NPV}_B = 40$; its cash arrives later, so it gains more from a low rate); for $r > r^*$, B does.

Answer (c). Indifferent at $r^* \approx 5.57\%$.

Step 5 — (d) Decision. At $r_0 = 11\% > r^*$, (7.3) exceeds (7.2): take project B. Both IRRs in (7.4) exceed 11%, so each project alone is acceptable; the choice between them follows the NPV. The IRR ranking happens to agree here, but it would rank wrongly at any cost of capital below 5.57%.

Answer (d). Take project B (higher NPV at 11%).

Intuition. IRR measures a rate, NPV measures value. When projects differ in timing, the one with earlier cash flows can have the higher IRR yet the lower NPV at low discount rates. The crossover rate is exactly the IRR of the difference between the projects.

Common mistakes.

- Choosing by IRR alone: correct here only because $11\% > r^*$.
- Finding the crossover by averaging the IRRs, or by equating the IRRs, instead of solving $\text{NPV}_A(r) = \text{NPV}_B(r)$.
- Taking both projects because both NPVs are positive: they are mutually exclusive.

Ref: Berk, §8.3 (alternative rules, IRR), pp. 216–226; §8.4 (choosing between projects, crossover), pp. 226–232.

8. Innovation Company: how many IRRs? — Berk 8.17, p. 243

Statement.

Source: Berk–DeMarzo–Harford, Ch. 8, Exercise 17, p. 243 (PDF 274) · Answer in book: no

Innovation Company is thinking about marketing a new software product. Upfront costs to market and develop the product are \$5 million. The product is expected to generate profits of \$1 million per year for ten years. The company will have to provide product support expected to cost \$100,000 per year in perpetuity. Assume all profits and expenses occur at the end of the year.

a. What is the NPV of this investment if the cost of capital is 6%? Should the firm undertake the project? Repeat the analysis for discount rates of 2% and 11%.

b. How many IRRs does this investment opportunity have?

c. What does the IRR rule indicate about this investment?

What is asked. (a) The NPV at 6%, 2% and 11%, with the accept/reject call at each rate. (b) How many discount rates set the NPV to zero. (c) Whether the "IRR above cost of capital" rule gives a usable answer for this project.

Approach. Value the cash flows with two standard formulas: a ten-year annuity for the profits and a perpetuity for the support cost. The cash flows change sign twice (−, +, −), so the NPV can cross zero twice, which is exactly when the IRR rule breaks down (Berk, "multiple IRRs").

Notation.

- $I_0 = 5$: up-front cost (\$ millions); $\pi = 1$: profit per year for years 1–10 (\$ millions).
- $s = 0.1$: support cost per year in perpetuity, from year 1 (\$ millions).
- $r > 0$: cost of capital; $a(r) = \big(1 - (1+r)^{-10}\big)/r$: ten-year annuity factor.
- $N(r)$: NPV of the project at rate r (\$ millions).

Step 1 — NPV function. The profits are an annuity of π for ten years, worth $\pi\, a(r)$; the support cost is a perpetuity of s starting at $t = 1$, worth s/r:

$$N(r) = -I_0 + \pi\, a(r) - \frac{s}{r}. \tag{8.1}$$

Step 2 — Annuity factors. Writing the formula $a(r) = \big(1 - (1+r)^{-10}\big)/r$, substituting each rate, then computing:

$$\begin{aligned}
a(0.06) &= \frac{1-(1.06)^{-10}}{0.06} = \frac{1-0.5583948}{0.06} = \frac{0.4416052}{0.06} = 7.36009,\\
a(0.02) &= \frac{1-(1.02)^{-10}}{0.02} = \frac{1-0.8203483}{0.02} = \frac{0.1796517}{0.02} = 8.98259,\\
a(0.11) &= \frac{1-(1.11)^{-10}}{0.11} = \frac{1-0.3521845}{0.11} = \frac{0.6478155}{0.11} = 5.88923.
\end{aligned}$$

Step 3 — (a) Evaluate at 6%, 2% and 11%. Substituting each rate and its factor into (8.1):

$$\begin{aligned}
N(0.06) &= -5 + (1)(7.36009) - \frac{0.1}{0.06}\\
&= -5 + 7.36009 - 1.66667\\
&= 0.69342, \qquad (8.2)\\
N(0.02) &= -5 + (1)(8.98259) - \frac{0.1}{0.02}\\
&= -5 + 8.98259 - 5.00000\\
&= -1.01741, \qquad (8.3)\\
N(0.11) &= -5 + (1)(5.88923) - \frac{0.1}{0.11}\\
&= -5 + 5.88923 - 0.90909\\
&= -0.01986. \qquad (8.4)
\end{aligned}$$

Undertake the project at 6% (NPV ≈ \$693,420); reject it at 2% and at 11%.

Answer (a). NPV = \$0.693M at 6% (accept); −\$1.017M at 2% and −\$0.020M at 11% (reject).

Step 4 — (b) Number of IRRs. For $r > 0$, multiply (8.1) by r (which preserves the zeros), substituting $a(r)$ and the data:

$$\begin{aligned}
r\,N(r) &= -5r + r \cdot \frac{1-(1+r)^{-10}}{r} - r \cdot \frac{0.1}{r}\\
&= -5r + \big(1 - (1+r)^{-10}\big) - 0.1\\
&= 0.9 - (1+r)^{-10} - 5r.
\end{aligned}$$

So $N(r) = 0$ if and only if

$$g(r) \equiv 0.9 - (1+r)^{-10} = 5r. \tag{8.5}$$

Since $(1+r)^{-10}$ is strictly convex, g is strictly concave, and a strictly concave function crosses a straight line at most twice. By (8.3)–(8.4), N changes sign on $(0.02, 0.06)$ and again on $(0.06, 0.11)$, so there are exactly two IRRs. Bracketing more tightly with (8.1): $N(0.027) = -0.0414$, $N(0.028) = 0.0465$, $N(0.108) = 0.0130$, $N(0.109) = -0.0034$. Narrowing each interval:

$$\text{IRR}_1 = 2.75\%, \qquad \text{IRR}_2 = 10.88\%. \tag{8.6}$$

Answer (b). Two IRRs: 2.75% and 10.88%.

Step 5 — (c) What the IRR rule says. The rule "accept if the IRR exceeds the cost of capital" is not defined when there are two IRRs: at $r = 6\%$ one IRR is above r and the other below. The NPV is positive exactly for $2.75\% < r < 10.88\%$ (by (8.5), between the two crossings) and negative outside, so only the NPV rule gives the right answer. At very low rates the perpetual support cost, s/r, dominates; at high rates the up-front cost does.

Answer (c). The IRR rule is uninformative (two IRRs); use NPV: accept only for $2.75\% < r < 10.88\%$.

Intuition. The cash flows change sign twice ($-$, then $+$ for ten years, then $-$ forever), so the project is partly an investment and partly a loan that is never repaid. Different rates favour different parts, and NPV is positive only in a window of rates.

Common mistakes.

- Netting the support cost into the profits for ten years only (0.9 per year) and forgetting that it continues forever after year 10.
- Assuming a lower discount rate always raises NPV: here a lower rate inflates the perpetual cost s/r faster than the ten-year profits.
- Reporting one IRR from a spreadsheet and applying the IRR rule to it.

Ref: Berk, §8.3 (IRR rule pitfalls: multiple IRRs), pp. 216–226.

9. CoffeeStop: WACC for the liquor division — Berk 13.20, p. 405

Statement.

Source: Berk–DeMarzo–Harford, Ch. 13, Exercise 20, p. 405 (PDF 436) · Answer in book: no

CoffeeStop primarily sells coffee. It recently introduced a premium coffee-flavored liquor. Suppose the firm faces a tax rate of 35% and collects the following information. If it plans to finance 11% of the new liquor-focused division with debt and the rest with equity, what WACC should it use for its liquor division? Assume a cost of debt of 4.8%, a risk-free rate of 3%, and a risk premium of 6%.

	Beta	% Equity	% Debt
CoffeeStop	0.61	96%	4%
BF Liquors	0.26	89%	11%

What is asked. Find the discount rate for CoffeeStop's new liquor division, financed 11% with debt, given betas and capital structures of CoffeeStop and of a liquor company.

Approach. Divisional (project-based) cost of capital: a division is discounted at the rate of a pure-play firm in the same business. BF Liquors is that comparable, and it already has the division's planned 11% debt, so its WACC (CAPM cost of equity, after-tax cost of debt) is the division's WACC, as in Berk's DuPont–Weyerhaeuser example. The unlever/relever route is shown as a check.

Notation.

- $\beta_E = 0.26$: equity beta of the pure-play comparable BF Liquors; $E/V = 0.89$, $D/V = 0.11$: its weights.
- $r_f = 0.03$; MRP $= 0.06$: market risk premium; $r_D = 0.048$: cost of debt; $\tau = 0.35$: tax rate.
- r_E: cost of equity; r_U: unlevered (pre-tax) cost of capital; r_{wacc}: after-tax WACC.

Step 1 — Choose the comparable. The division sells liquor, so CoffeeStop's own beta (coffee) is the wrong risk. BF Liquors is the pure play, and it has the same leverage, $D/V = 0.11$, that is planned for the division. Its WACC is therefore the division's WACC (Berk, divisional costs of capital).

Step 2 — Cost of equity (CAPM). Substituting the comparable's beta into the security market line:

$$\begin{aligned} r_E &= r_f + \beta_E \,\text{MRP} \\ &= (0.03) + (0.26)(0.06) \\ &= 0.03 + 0.0156 \\ &= 0.0456. \end{aligned} \tag{9.1}$$

Step 3 — WACC. The after-tax WACC weights each cost by its share of value, debt net of the tax saving:

$$r_{\text{wacc}} = \frac{E}{V} r_E + \frac{D}{V} r_D (1 - \tau). \tag{9.2}$$

Substituting (9.1) and the data; first $r_D(1-\tau) = (0.048)(0.65) = 0.0312$:

$$\begin{aligned} r_{\text{wacc}} &= (0.89)(0.0456) + (0.11)(0.048)(1 - 0.35) \\ &= (0.89)(0.0456) + (0.11)(0.0312) \\ &= 0.040584 + 0.003432 \\ &= 0.044016 \approx 4.40\%. \end{aligned} \tag{9.3}$$

Answer (WACC). $r_{\text{wacc}} \approx 4.40\%$.

Step 4 — Check: unlever and relever. The pre-tax WACC equals the unlevered cost of capital (as proved in Exercise 4):

$$\begin{aligned} r_U &= \frac{E}{V} r_E + \frac{D}{V} r_D \\ &= (0.89)(0.0456) + (0.11)(0.048) \\ &= 0.040584 + 0.00528 \\ &= 0.045864. \end{aligned} \tag{9.4}$$

Subtracting the definition of r_U from (9.2) leaves only the tax term, $r_{\text{wacc}} - r_U = \frac{D}{V} r_D(1-\tau) - \frac{D}{V} r_D = -\frac{D}{V}\tau r_D$, so $r_{\text{wacc}} = r_U - \frac{D}{V}\tau r_D$. Substituting (9.4) at the division's $D/V = 0.11$:

$$\begin{aligned} r_{\text{wacc}} &= (0.045864) - (0.11)(0.35)(0.048) \\ &= 0.045864 - 0.001848 \\ &= 0.044016, \end{aligned}$$

the same as (9.3), as it must be when the division copies the comparable's leverage.

Remark. The data are not fully consistent under the CAPM: the cost of debt implies a debt beta $(0.048 - 0.03)/0.06 = 0.30$, above the equity beta of 0.26, so $r_E < r_D$. The answer uses the inputs as given. Using CoffeeStop's own WACC, $(0.96)(0.03 + 0.61 \cdot 0.06) + (0.04)(0.048)(0.65) = (0.96)(0.0666) + 0.001248 = 0.063936 + 0.001248 = 0.065184$, would overstate the division's rate by about 2.1 points.

Intuition. A firm should discount each line of business at the rate investors demand for that business. A liquor division is valued like a liquor company, not like the parent.

Common mistakes.

- Using CoffeeStop's beta 0.61 (or its WACC, 6.5%) for the liquor division.
- Forgetting the $(1-\tau)$ on the cost of debt, which gives the pre-tax 4.59%.
- Unlevering BF Liquors' beta and then relevering at CoffeeStop's 4% debt instead of the division's planned 11%.

Ref: Berk, §13.3 (WACC), p. 390; §13.5 (project-based costs of capital, divisional costs), pp. 396–398.

10. Karamazov Semiconductors: enterprise vs. equity IRR — Higgins 8.16, p. 330

Statement.

Source: Higgins (Koski–Higgins–Mitton), Ch. 8, Exercise 16, p. 330 (PDF 351) · Answer in book: no

Karamazov Semiconductors is considering an investment to expand its existing line of business. The investment will cost \$10 million and is expected to produce after-tax cash flows of \$1 million per year in perpetuity. Karamazov's capital structure is composed of 50 percent debt and 50 percent equity, and it plans to finance the investment using this same ratio of financing. Karamazov's pre-tax cost of debt is 8 percent, its tax rate is 25 percent, and its cost of equity is 14 percent.

a. What is Karamazov's weighted-average cost of capital?

b. Calculate the IRR of the project from the enterprise perspective.

c. Calculate the IRR of the project from the equity perspective.

d. Explain the difference between your two answers in parts (b) and (c).

What is asked. (a) The firm's WACC from the 50/50 structure and the costs given. (b) The project's IRR on the whole investment (enterprise view). (c) The IRR on the shareholders' own money, after paying the lenders (equity view). (d) Why (b) and (c) differ and how each is to be used.

Approach. Higgins' enterprise and equity perspectives (§8.6.1): the enterprise view discounts unlevered cash flows at the WACC; the equity view nets out debt flows and uses the cost of equity. A perpetuity's IRR is simply its annual cash flow over the investment, because $-I + C/\text{IRR} = 0$.

Notation.

- $I = 10$: investment (\$ millions); $C = 1$: after-tax (unlevered) cash flow per year in perpetuity (\$ millions).
- $D/V = E/V = 0.5$: financing weights; $D = 5$, $E = 5$: debt and equity raised (\$ millions).
- $K_D = 0.08$: pre-tax cost of debt; $t = 0.25$: tax rate; $K_E = 0.14$: cost of equity; K_W: WACC.
- Assumption: the debt is perpetual (kept at 50% of the project), interest paid annually.

Step 1 — (a) WACC. Weighted average of the after-tax cost of debt and the cost of equity:

$$\begin{aligned} K_W &= \frac{D}{V}(1-t)K_D + \frac{E}{V}K_E \\ &= (0.5)(1-0.25)(0.08) + (0.5)(0.14) \\ &= (0.5)(0.06) + 0.07 \\ &= 0.03 + 0.07 \\ &= 0.10. \end{aligned} \tag{10.1}$$

Answer (a). $K_W = 10\%$.

Step 2 — (b) Enterprise IRR. Enterprise cash flows are $-I$ at $t = 0$ and C forever. A perpetuity of C discounted at rate ρ is worth C/ρ, so the IRR solves $-I + C/\text{IRR} = 0$; multiplying by IRR and dividing by I:

$$\begin{aligned} \text{IRR}_{\text{ent}} &= \frac{C}{I} = \frac{(1)}{(10)} \\ &= 0.10. \end{aligned} \tag{10.2}$$

Answer (b). $\text{IRR}_{\text{ent}} = 10\%$.

Step 3 — (c) Equity IRR. Equity puts up $E = 5$; lenders put up $D = 5$ and receive interest $K_D D = (0.08)(5) = 0.4$ a year. Interest is deductible, saving $t\,K_D D = (0.25)(0.4) = 0.1$ of tax, so the after-tax cost to the shareholders is

$$\begin{aligned} K_D(1-t)D &= (0.08)(1-0.25)(5) \\ &= 0.4 - 0.1 = 0.3, \end{aligned} \tag{10.3}$$

$$\begin{aligned} C_E &= C - K_D(1-t)D = (1) - (0.3) \\ &= 0.7, \end{aligned} \tag{10.4}$$

$$\begin{aligned} \text{IRR}_{\text{eq}} &= \frac{C_E}{E} = \frac{(0.7)}{(5)} \\ &= 0.14. \end{aligned} \tag{10.5}$$

(The perpetuity argument of Step 2 applies to the equity flows -5, then 0.7 forever.)

Answer (c). $\text{IRR}_{\text{eq}} = 14\%$.

Step 4 — (d) Why they differ. The two IRRs measure returns on different cash flows: the enterprise IRR (10.2) is the return on all capital, the equity IRR (10.5) is the return on the shareholders' part only, after paying debt holders. Leverage concentrates the surplus over the after-tax cost of debt on a smaller equity base, so the equity IRR is higher and also riskier. Each IRR must be compared with its own hurdle: 10% against $K_W = 10\%$ and 14% against $K_E = 14\%$. Both comparisons give the same verdict: the project earns exactly its cost of capital (NPV $= -10 + 1/0.10 = -10 + 10 = 0$). Comparing the equity IRR with the WACC (or the enterprise IRR with the cost of equity) would mix perspectives.

Answer (d). Different cash flows and risk; compare 10% with K_W and 14% with K_E: both say NPV $= 0$.

Intuition. Debt does not create the extra 4 points of equity return; it transfers risk. Viewed consistently, the enterprise and equity perspectives always agree on whether a project creates value.

Common mistakes.

- Comparing the 14% equity IRR with the 10% WACC and concluding the project is great.
- Deducting pre-tax interest (0.4) instead of after-tax interest (0.3) in the equity flows.
- Using the full \$10 million as the equity investment in (c).

Ref: Higgins, §8.6.1 (the enterprise perspective vs. the equity perspective), pp. 307–310.

11. Percolated Fiber: fixing the consultants' report — Berk 9.25, p. 277

Statement.

Source: Berk–DeMarzo–Harford, Ch. 9, Exercise 25, p. 277 (PDF 308) · Answer in book: no

You are a manager at Percolated Fiber, which is considering expanding its operations in synthetic fiber manufacturing. Your boss comes into your office, drops a consultant's report on your desk, and complains,

"We owe these consultants $1 million for this report, and I am not sure their analysis makes sense. Before we spend the $25 million on new equipment needed for this project, look it over and give me your opinion." You open the report and find the following estimates (in thousands of dollars):

Year	1	2	...	9	10
Sales Revenue	30,000	30,000		30,000	30,000
Costs of Goods Sold	18,000	18,000		18,000	18,000
Gross Profit	12,000	12,000		12,000	12,000
General, Sales, and Administrative Expenses	2,000	2,000		2,000	2,000
Depreciation	2,500	2,500		2,500	2,500
EBIT	7,500	7,500		7,500	7,500
Income Tax	2,625	2,625		2,625	2,625
Net Income	4,875	4,875		4,875	4,875

All of the estimates in the report seem correct. You note that the consultants used straight-line depreciation for the new equipment that will be purchased today (year 0), which is what the accounting department recommended. They also calculated the depreciation assuming no salvage value for the equipment, which is the company's assumption in this case. The report concludes that because the project will increase earnings by $4.875 million per year for ten years, the project is worth $48.75 million. You think back to your glory days in finance class and realize there is more work to be done!

First, you note that the consultants have not factored in the fact that the project will require $10 million in working capital upfront (year 0), which will be fully recovered in year 10. Next, you see they have attributed $2 million of selling, general, and administrative expenses to the project, but you know that $1 million of this amount is overhead that will be incurred even if the project is not accepted. Finally, you know that accounting earnings are not the right thing to focus on!

a. Given the available information, what are the free cash flows in years 0 through 10 that should be used to evaluate the proposed project?

b. If the cost of capital for this project is 14%, what is your estimate of the value of the new project?

What is asked. (a) The correct free cash flows in years 0 to 10, repairing the consultants' earnings forecast (missing working capital, non-incremental overhead, earnings instead of cash). (b) The project's value (NPV) at a 14% cost of capital.

Approach. Free cash flow = unlevered net income + depreciation − capex − increase in NWC (Berk §9.3). Only incremental items count, so allocated overhead and the consultants' fee are excluded. The years 1–9 flows are equal, so the ten-year annuity formula values them, with the NWC recovery discounted separately.

Notation.

- Amounts in thousands of dollars; $t = 0, \ldots, 10$ years.
- $K = 25{,}000$: capital expenditure at $t = 0$; Dep = 2,500: straight-line depreciation, years 1–10.
- NWC = 10,000: working capital invested at $t = 0$, recovered at $t = 10$.
- SGA = 1,000: incremental SG&A (the other 1,000 is overhead incurred anyway).
- τ: tax rate; FCF_t: free cash flow; $r = 0.14$: cost of capital; a: ten-year annuity factor at 14%.

Step 1 — Tax rate. The report's tax line over its EBIT line gives the rate: $\tau = 2{,}625/7{,}500 = 0.35$. Depreciation $25{,}000/10 = 2{,}500$ matches the report.

Step 2 — Irrelevant items. The $1 million consultants' fee is owed whatever is decided (sunk): excluded. Only the incremental 1,000 of SG&A enters.

Step 3 — Incremental EBIT and unlevered net income (years 1–10). Revenue minus

incremental costs and depreciation, then tax at 35%:

$$\begin{aligned} \text{EBIT} &= 30{,}000 - 18{,}000 - (1{,}000) - 2{,}500 \\ &= 8{,}500, \\ \text{Tax} &= (0.35)(8{,}500) = 2{,}975, \\ \text{Unlevered NI} &= 8{,}500 - (2{,}975) = 5{,}525. \end{aligned} \tag{11.1}$$

Step 4 — (a) Free cash flows. $\text{FCF}_t = \text{Unlevered NI}_t + \text{Dep}_t - \text{CapEx}_t - \Delta\text{NWC}_t$. At $t = 0$ only capex and the NWC build-up occur; in years 1–9 add back depreciation (a non-cash charge); in year 10 the NWC comes back ($\Delta\text{NWC}_{10} = -10{,}000$):

$$\begin{aligned} \text{FCF}_0 &= -(25{,}000) - (10{,}000) = -35{,}000, \\ \text{FCF}_{1\ldots9} &= (5{,}525) + (2{,}500) = 8{,}025, \\ \text{FCF}_{10} &= (8{,}025) + (10{,}000) = 18{,}025. \end{aligned} \tag{11.2}$$

Cross-check with the tax-shield form, $\text{FCF} = (\text{Rev} - \text{Costs})(1-\tau) + \tau\,\text{Dep}$: $(30{,}000 - 18{,}000 - 1{,}000)(0.65) + (0.35)(2{,}500) = (11{,}000)(0.65) + 875 = 7{,}150 + 875 = 8{,}025$.

Answer (a). $\text{FCF}_0 = -35{,}000$; $\text{FCF}_{1\ldots9} = 8{,}025$; $\text{FCF}_{10} = 18{,}025$ ($ thousands).

Step 5 — Annuity factor and year-10 factor. Writing the formula, substituting $r = 0.14$, $T = 10$, then computing:

$$(1.14)^{10} = 3.707221, \qquad (1.14)^{-10} = 0.2697438,$$

$$a = \frac{1 - (1+r)^{-T}}{r} = \frac{1 - 0.2697438}{0.14} = \frac{0.7302562}{0.14} = 5.216116.$$

Step 6 — (b) Value at 14%. Writing FCF_{10} as 8,025 plus the NWC recovery, so that 8,025 is a ten-year annuity:

$$\begin{aligned} \text{NPV} &= -35{,}000 + 8{,}025\,a + \frac{10{,}000}{(1.14)^{10}} \\ &= -35{,}000 + (8{,}025)(5.216116) + \frac{10{,}000}{(3.707221)} \\ &= -35{,}000 + 41{,}859.33 + 2{,}697.44 \\ &= 9{,}556.77. \end{aligned} \tag{11.3}$$

The project adds about $9.56 million of value (the future cash flows are worth $44.56 million against $35 million invested), far from the consultants' $48.75 million.

Answer (b). NPV ≈ $9,556.77 thousand (≈ $9.56 million).

Intuition. Earnings are not cash: the equipment is paid for up front, not in yearly instalments of depreciation; working capital ties up cash until year 10; and money in year 10 is worth much less than money today. Overhead that exists anyway is not caused by the project.

Common mistakes.

- Keeping the full $2 million SG&A (the $1 million overhead is not incremental), or subtracting the $1 million consultants' fee (sunk).
- Forgetting to add back depreciation, or subtracting the $25 million capex again in each year through depreciation *and* at $t = 0$.
- Omitting the NWC recovery in year 10, or the NWC outflow at $t = 0$.

Ref: Berk, §9.2–9.3 (incremental earnings and free cash flow), pp. 250–260; §9.4 (NWC, sunk costs, overhead), pp. 260–265.

12. Torino Marine: three terminal values — Higgins 9.11, p. 382

Statement.

Source: Higgins (Koski–Higgins–Mitton), Ch. 9, Exercise 11, p. 382 (PDF 403) · Answer in book: yes, p. 427

The following is a four-year forecast for Torino Marine.

Year	2022	2023	2024	2025
Free cash flow ($ millions)	–52	76	92	112

a. Estimate the fair market value of Torino Marine at the end of 2021. Assume that after 2025, earnings before interest and tax will remain constant at $200 million, depreciation will equal capital expenditures in each year, and working capital will not change. Torino Marine's weighted-average cost of capital is 11 percent and its tax rate is 40 percent.

b. Estimate the fair market value per share of Torino Marine's equity at the end of 2021 if the company has 40 million shares outstanding and the market value of its interest-bearing liabilities on the valuation date equals $250 million.

c. Now let's try a different terminal value. Estimate the fair market value of Torino Marine's equity per share at the end of 2021 under the following assumptions:

   i. Free cash flows in years 2022 through 2025 remain as above.

   ii. EBIT in year 2025 is $200 million, and then grows at 5 percent per year forever.

   iii. To support the perpetual growth in EBIT, capital expenditures in year 2026 exceed depreciation by $30 million, and this difference grows 5 percent per year forever.

   iv. Similarly, working capital investments are $15 million in 2026, and this amount grows 5 percent per year forever.

d. Lastly, let's try a third terminal value. Estimate the fair market value of Torino Marine's equity per share at the end of 2021 under the following assumptions:

   i. Free cash flows in years 2022 through 2025 remain as above. EBIT in year 2025 will be $200 million.

   ii. At year-end 2025, Torino Marine has reached maturity, and its equity sells for a "typical" multiple of year 2025 net income. Use 12 as a typical multiple.

   iii. At year-end 2025, Torino Marine has $250 million of interest-bearing liabilities outstanding at an average interest rate of 10 percent.

What is asked. (a) The firm (enterprise) value at end-2021 with a no-growth terminal value after 2025. (b) The equity value per share implied by (a). (c) Equity per share when the terminal value assumes 5% perpetual growth with the reinvestment it needs. (d) Equity per share when the terminal value comes from a P/E multiple of 12 on 2025 net income.

Approach. Discounted free cash flow (Higgins §9.2): enterprise value = PV of the explicit FCF forecast + PV of a terminal value at 2025, all at the WACC; equity = enterprise value − interest-bearing debt. The three parts change only the terminal value: a level perpetuity, a growing perpetuity, and a market multiple (which values equity, so debt is added back).

Notation.

- Amounts in $ millions; valuation date end of 2021; $\mathrm{FCF}_{2022}, \ldots, \mathrm{FCF}_{2025} = -52, 76, 92, 112$.
- $K_W = 0.11$: WACC; $t = 0.40$: tax rate; EBIT: earnings before interest and tax.
- TV_{2025}: terminal (enterprise) value at end of 2025; FMV: enterprise value at end of 2021.
- $D = 250$: interest-bearing liabilities; $n = 40$: shares (millions); $g = 0.05$: growth in (c).

Step 1 — PV of the explicit forecast (common to all parts). Year 2022 is one year after the valuation date, 2025 four years; the factors are 1.11, $(1.11)^2 = 1.2321$, $(1.11)^3 = 1.367631$, $(1.11)^4 = 1.518070$:

$$\begin{aligned} PV_{FCF} &= \frac{-52}{(1.11)} + \frac{76}{(1.2321)} + \frac{92}{(1.367631)} + \frac{112}{(1.518070)} \\ &= -46.847 + 61.683 + 67.270 + 73.778 \\ &= 155.884. \end{aligned} \tag{12.1}$$

Any terminal value dated 2025 is divided by $(1.11)^4 = 1.518070$.

Step 2 — (a) No-growth terminal value. With depreciation equal to capex and no change in working capital, $\text{FCF} = \text{EBIT}(1-t) + \text{Dep} - \text{CapEx} - \Delta\text{NWC} = \text{EBIT}(1-t)$ from 2026 on, a level perpetuity valued at FCF/K_W:

$$\begin{aligned} TV_{2025} &= \frac{\text{EBIT}(1-t)}{K_W} = \frac{(200)(1-0.40)}{(0.11)} \\ &= \frac{120}{0.11} = 1{,}090.909, \\ PV(TV) &= \frac{1{,}090.909}{(1.518070)} = 718.616, \\ \text{FMV}_a &= (155.884) + (718.616) = 874.500. \end{aligned} \tag{12.2}$$

Answer (a). FMV ≈ \$874.5 million.

Step 3 — (b) Equity per share. Equity is enterprise value minus debt, divided by shares. Substituting (12.2):

$$\begin{aligned} \frac{\text{FMV}_a - D}{n} &= \frac{(874.500) - (250)}{(40)} \\ &= \frac{624.500}{40} = 15.61 \text{ dollars per share.} \end{aligned} \tag{12.3}$$

Answer (b). \$15.61 per share.

Step 4 — (c) Growing terminal value. 2026 EBIT is 200(1.05); FCF subtracts net capex (30) and working-capital investment (15). All three pieces grow at 5%, so FCF grows at 5% and the terminal value is the growing perpetuity $\text{FCF}_{2026}/(K_W - g)$:

$$\begin{aligned} \text{FCF}_{2026} &= (200)(1.05)(1-0.40) - 30 - 15 \\ &= (210)(0.60) - 45 \\ &= 126 - 45 = 81, \\ TV_{2025} &= \frac{(81)}{(0.11) - (0.05)} = \frac{81}{0.06} = 1{,}350, \\ PV(TV) &= \frac{1{,}350}{(1.518070)} = 889.287, \\ \text{FMV}_c &= (155.884) + (889.287) = 1{,}045.171, \\ \text{per share} &= \frac{(1{,}045.171) - (250)}{(40)} = \frac{795.171}{40} = 19.88. \end{aligned} \tag{12.4}$$

Answer (c). \$19.88 per share (enterprise value \$1,045.2 million).

Step 5 — (d) Multiple-based terminal value. 2025 net income deducts interest (10% on 250) before tax; the multiple then values equity, and debt is added back to get an enterprise value:

$$\begin{aligned} NI_{2025} &= (200 - (0.10)(250))(1 - 0.40) \\ &= (200 - 25)(0.60) = (175)(0.60) = 105, \\ E_{2025} &= (12)(105) = 1{,}260, \\ TV_{2025} &= E_{2025} + D = (1{,}260) + (250) = 1{,}510, \\ PV(TV) &= \frac{1{,}510}{(1.518070)} = 994.684, \\ \text{FMV}_d &= (155.884) + (994.684) = 1{,}150.568, \\ \text{per share} &= \frac{(1{,}150.568) - (250)}{(40)} = \frac{900.568}{40} = 22.51. \end{aligned} \tag{12.5}$$

The P/E multiple values equity, so debt is added back to get an enterprise terminal value that can be discounted at the WACC together with the FCFs.

Answer (d). $22.51 per share.

Intuition. Most of the value (82–86%) sits in the terminal value, so the method chosen for it drives the answer: $15.61, $19.88 or $22.51 per share. Growth is worth something only because the reinvestment it needs ($45 million in 2026) earns more than the WACC; the market multiple of 12 implies even more optimism than 5% growth.

Common mistakes.

- Discounting the 2025 terminal value by $(1.11)^5$, or treating it as a 2026 amount.
- Using EBIT instead of EBIT$(1 - t)$ as the perpetual FCF, or forgetting net capex and NWC in the growth case.
- In (d), discounting the P/E equity value as if it were enterprise value (forgetting to add back the $250 million), or applying the multiple to EBIT.

Book answer. Matches the book's answers (p. 427): (a) $874.5 million, (b) $15.61, (c) $1,045.2 million and $19.88, (d) $22.51 per share. In (d) the book adds $155.8 instead of $155.9 for the forecast-period PV and prints $1,150.5 million; carried to three decimals the enterprise value is $1,150.568 million. The per-share value is unaffected.

Ref: Higgins, §9.2.1–9.2.3 (free cash flow, the terminal value, a numerical example), pp. 342–348; §9.3 (valuation by comparables, P/E), p. 349.

13. Calculating project NPV (CUTO) — Ross 10.31 (Desafio), p. 356

Statement.

Source: Ross, Cap. 10, Exercise 31 (DESAFIO), p. 356 (PDF 380) · Answer in book: no

Você foi contratado como consultor da Cítaras Urbanas De Tecnologia Original Ltda. (CUTO), fabricante de cítaras de qualidade. O mercado de cítaras está crescendo rapidamente. A empresa comprou terras há três anos por $1,4 milhão, prevendo o seu uso como depósito de lixo tóxico, mas contratou outra empresa para lidar com todo o material tóxico. Com base em uma avaliação recente, a empresa acredita que poderia vender o terreno por $1,5 milhão após impostos. Em quatro anos, o terreno poderia ser vendido por $1,6 milhão após impostos. A empresa também contratou uma empresa de marketing para analisar o mercado de cítaras a um custo de $125.000. Este é um trecho extraído do relatório de marketing:

> A indústria de cítaras terá uma expansão rápida nos próximos quatro anos. Com o reconhecimento de marca que a CUTO tem, cremos que a empresa poderá vender 3.200, 4.300, 3.900 e 2.800 unidades por ano nos próximos quatro anos, respectivamente. Mais uma vez, aproveitando o reconhecimento da CUTO, acreditamos que um preço-prêmio de $780 por cítara possa ser cobrado.

> Como as cítaras parecem ser uma moda passageira, é provável que, ao final do período de quatro anos, as vendas sejam interrompidas.

A CUTO acredita que os custos fixos do projeto serão de \$425.000 por ano, e os custos variáveis serão de 15% das vendas. O equipamento necessário para a produção custará \$4,2 milhões e será depreciado de forma linear e acelerada, em três anos, aproveitando uma depreciação incentivada combinada com a depreciação fiscal. Ao final do projeto, o equipamento pode ser sucateado por \$400.000. Um capital de giro de \$125.000 é necessário imediatamente para financiar o investimento em ativos circulantes. A CUTO tem uma alíquota tributária de 34% e o retorno exigido sobre o projeto é de 13%. Qual é o VPL do projeto? Assuma que a empresa tenha outros projetos lucrativos.

What is asked. Find the NPV at 13% of CUTO's four-year zither project, deciding how to treat the land (bought for \$1.4 million, worth \$1.5 million now and \$1.6 million in four years), the marketing study, depreciation, salvage of the equipment and working capital.

Approach. Build incremental cash flows year by year (Ross §10.2–10.4): operating cash flow = EBIT$(1-\tau)$+ depreciation; add capital spending, NWC and after-tax salvage; then discount. The land enters at its opportunity cost (today's after-tax sale value) and comes back at its after-tax value in year 4; the land's purchase price and the study are sunk.

Notation.

- Amounts in dollars; $t = 0, \ldots, 4$ years; $\tau = 0.34$: tax rate; $R = 0.13$: required return.
- $L_0 = 1{,}500{,}000$: after-tax value of the land today (opportunity cost); $L_4 = 1{,}600{,}000$: after-tax land value at $t = 4$.
- q_t: units sold (3,200, 4,300, 3,900, 2,800); $P = 780$: price; $S_t = Pq_t$: sales.
- $\text{VC}_t = 0.15\, S_t$: variable cost; $\text{FC} = 425{,}000$: fixed cost per year.
- $K = 4{,}200{,}000$: equipment; Dep_t: tax depreciation; $SV = 400{,}000$: salvage price at $t = 4$.
- $\text{NWC} = 125{,}000$: working capital at $t = 0$; OCF_t: operating cash flow; CF_t: total cash flow.

Step 1 — Relevant and irrelevant items. The \$1.4 million paid for the land three years ago and the \$125,000 marketing study are sunk: excluded. Using the land forgoes selling it now for L_0, so L_0 is a $t = 0$ outflow; at the end the land can still be sold for L_4, a $t = 4$ inflow. "Assume the firm has other profitable projects" means any tax loss could be used against other income; it does not bind here because EBIT is positive in every year (Step 3).

Step 2 — Depreciation. The equipment is "depreciado de forma linear e acelerada, em três anos": straight line to zero over three years, as in Ross's Problem 9 of the same chapter (p. 353):

$$\text{Dep}_{1,2,3} = \frac{(4{,}200{,}000)}{3} = 1{,}400{,}000, \qquad \text{Dep}_4 = 0. \tag{13.1}$$

The book value at $t = 4$ is therefore $4{,}200{,}000 - 3(1{,}400{,}000) = 0$.

Step 3 — Operating cash flow. For each year, $\text{EBIT}_t = S_t - \text{VC}_t - \text{FC} - \text{Dep}_t$ and $\text{OCF}_t = \text{EBIT}_t(1-\tau) + \text{Dep}_t$ (tax is paid on EBIT; depreciation is a non-cash charge added back). Year 1 in full:

$$\begin{aligned}
S_1 &= (3{,}200)(780) = 2{,}496{,}000,\\
\text{VC}_1 &= (0.15)(2{,}496{,}000) = 374{,}400,\\
\text{EBIT}_1 &= 2{,}496{,}000 - (374{,}400) - (425{,}000) - (1{,}400{,}000) = 296{,}600,\\
\text{Tax}_1 &= (0.34)(296{,}600) = 100{,}844,\\
NI_1 &= 296{,}600 - (100{,}844) = 195{,}756,\\
\text{OCF}_1 &= (195{,}756) + (1{,}400{,}000) = 1{,}595{,}756.
\end{aligned} \tag{13.2}$$

Years 2–4, the same lines:

$$
\begin{aligned}
S_2 &= (4{,}300)(780) = 3{,}354{,}000, \quad \mathrm{VC}_2 = (0.15)(3{,}354{,}000) = 503{,}100,\\
\mathrm{EBIT}_2 &= 3{,}354{,}000 - 503{,}100 - 425{,}000 - 1{,}400{,}000 = 1{,}025{,}900,\\
\mathrm{OCF}_2 &= 1{,}025{,}900 - (0.34)(1{,}025{,}900) + 1{,}400{,}000 = 1{,}025{,}900 - 348{,}806 + 1{,}400{,}000 = 2{,}077{,}094,\\
S_3 &= (3{,}900)(780) = 3{,}042{,}000, \quad \mathrm{VC}_3 = (0.15)(3{,}042{,}000) = 456{,}300,\\
\mathrm{EBIT}_3 &= 3{,}042{,}000 - 456{,}300 - 425{,}000 - 1{,}400{,}000 = 760{,}700,\\
\mathrm{OCF}_3 &= 760{,}700 - (0.34)(760{,}700) + 1{,}400{,}000 = 760{,}700 - 258{,}638 + 1{,}400{,}000 = 1{,}902{,}062,\\
S_4 &= (2{,}800)(780) = 2{,}184{,}000, \quad \mathrm{VC}_4 = (0.15)(2{,}184{,}000) = 327{,}600,\\
\mathrm{EBIT}_4 &= 2{,}184{,}000 - 327{,}600 - 425{,}000 - 0 = 1{,}431{,}400,\\
\mathrm{OCF}_4 &= 1{,}431{,}400 - (0.34)(1{,}431{,}400) + 0 = 1{,}431{,}400 - 486{,}676 = 944{,}724.
\end{aligned}
$$

Summary:

Year	1	2	3	4
Sales S_t	2,496,000	3,354,000	3,042,000	2,184,000
Variable cost (15%)	374,400	503,100	456,300	327,600
Fixed cost	425,000	425,000	425,000	425,000
Depreciation	1,400,000	1,400,000	1,400,000	0
EBIT	296,600	1,025,900	760,700	1,431,400
Tax (34%)	100,844	348,806	258,638	486,676
Net income	195,756	677,094	502,062	944,724
OCF	1,595,756	2,077,094	1,902,062	944,724

Step 4 — After-tax salvage. The equipment is sold for SV with book value zero, so the whole price is a taxable gain:

$$
\begin{aligned}
SV(1-\tau) &= 400{,}000 - (0.34)(400{,}000 - 0)\\
&= 400{,}000 - 136{,}000 = 264{,}000.
\end{aligned}
\tag{13.3}
$$

Step 5 — Total cash flows. Adding capital items to the OCFs, assuming the working capital is fully recovered when sales stop:

$$
\begin{aligned}
\mathrm{CF}_0 &= -(4{,}200{,}000) - (1{,}500{,}000) - (125{,}000) = -5{,}825{,}000,\\
\mathrm{CF}_{1,2,3} &= \mathrm{OCF}_{1,2,3} = 1{,}595{,}756,\ 2{,}077{,}094,\ 1{,}902{,}062,\\
\mathrm{CF}_4 &= (944{,}724) + (1{,}600{,}000) + (264{,}000) + (125{,}000) = 2{,}933{,}724.
\end{aligned}
\tag{13.4}
$$

Step 6 — NPV at 13%. The discount factors are 1.13, $(1.13)^2 = 1.2769$, $(1.13)^3 = 1.442897$, $(1.13)^4 = 1.63047361$. Discounting (13.4):

$$
\begin{aligned}
\mathrm{NPV} &= -5{,}825{,}000 + \frac{1{,}595{,}756}{(1.13)} + \frac{2{,}077{,}094}{(1.2769)} + \frac{1{,}902{,}062}{(1.442897)} + \frac{2{,}933{,}724}{(1.63047361)}\\
&= -5{,}825{,}000 + 1{,}412{,}173.45 + 1{,}626{,}669.28 + 1{,}318{,}224.38 + 1{,}799{,}307.87\\
&= 331{,}374.98.
\end{aligned}
\tag{13.5}
$$

The NPV is positive: CUTO should go ahead.

Answer (NPV). NPV $\approx$ \$331,375 > 0: accept.

Remark. The English original of this problem uses the U.S. three-year MACRS schedule (33.33%, 44.45%, 14.81%, 7.41%). With that schedule (book value also zero at $t = 4$) the NPV is \$337,231.54: the faster write-off brings tax savings forward, but the decision is unchanged.

Intuition. The land is not free because the firm already owns it: giving up \$1.5 million today to get \$1.6 million back in four years costs the project $1{,}500{,}000 - 1{,}600{,}000/1.63047361 = 1{,}500{,}000 - 981{,}309.96 = 518{,}690.04$, about \$519,000 in present value at 13%. The project still pays for that and for the equipment, so it creates value.

Common mistakes.

- Charging the \$1.4 million historical land cost or the \$125,000 study: both are sunk.
- Ignoring the land's year-4 resale value of \$1.6 million (or its $t = 0$ opportunity cost).
- Adding the full \$400,000 salvage without taxing the gain over the zero book value, or forgetting to recover the \$125,000 NWC in year 4.

Ref: Ross, §10.2 (incremental cash flows: sunk and opportunity costs, NWC), pp. 320–322; §10.3–10.4 (pro forma statements, depreciation, salvage), pp. 322–339.

14. State prices from a traded project — Amaro de Matos 1.2.6 #5, p. 28

Statement.

Source: Amaro de Matos, Ch. 1, Exercise 1.2.6 #5, p. 28 (PDF 46) · Answer in book: yes, p. 247

Consider a one-period economy with two possible states of nature at time $t = 1$. The risk-free rate in this economy is 1%. There is a project (call it project 1) in this economy with value today of 6.7 and payoffs $\Gamma_{11} = 10$ and $\Gamma_{12} = 5$.

(a) Compute the vector of state-prices when there are no arbitrage opportunities.

(b) Consider another project (project 2) with payoffs $\Gamma_{21} = -2$ and $\Gamma_{22} = 4$. The investment necessary to do project 1 is 5; to do project 2, we must invest 2. Will an agent invest in both projects?

Context needed to solve it *From Amaro de Matos §1.2.1 "One-Period Model", pp. 18–19 (PDF 36–37):*

Uncertainty is thus described by a set of states belonging to a finite sample space $\Omega = \{w_1, w_2, \ldots, w_K\}$, where each state w_j occurs with probability $p(w_j)$, denoted for simplicity as p_j. [...] Also, instead of having one single project to consider for investment, assume that the investor faces a bundle of N different one-period projects. Each of these projects has an initial value V_i, $i = 1, \ldots, N$, and at time $t = 1$ pays $\Gamma_i(w_j)$ in state $j = 1, 2, \ldots, K$. For simplicity of notation, let $\Gamma_i(w_j)$ be denoted by Γ_{ij}. These values define a value vector $V \in R^N$ and a payoff $N \times K$ matrix Γ, given by

$$\Gamma = (\Gamma_1, \Gamma_2, \ldots, \Gamma_K) = \begin{bmatrix} \Gamma_{11} & \Gamma_{12} & \cdots & \Gamma_{1K} \\ \Gamma_{21} & \Gamma_{22} & \cdots & \Gamma_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ \Gamma_{N1} & \Gamma_{N2} & \cdots & \Gamma_{NK} \end{bmatrix}$$

where Γ_j denotes the N-dimensional column vector of payoffs in state j.

From Amaro de Matos §1.2.1, p. 21 (PDF 39):

In other words, completeness of the market is equivalent to having the number of projects with linearly independent payoffs N equal to the number of states K. [...] In other words, an optimal portfolio choice links the N-dimensional value vector of the projects V to the $N \times K$ matrix of future payoffs by the relationship

$$V = \sum_{j=1}^{K} p_j \Gamma_j \frac{\partial U/\partial y_j}{\partial U/\partial x} = \sum_{j=1}^{K} \Gamma_j \psi_j,$$

where

$$p_j \frac{\partial U/\partial y_j}{\partial U/\partial x} \equiv \psi_j > 0 \qquad (1.12)$$

given the assumption that utilities are strictly increasing. [...] For that reason the K-dimensional vector ψ is termed the vector of *state-prices* and ψ_j is called the price of state j.

From Amaro de Matos §1.2.2 "Value and the Absence of Arbitrage", pp. 22–23 (PDF 40–41):

Proposition 1.2.1. *The absence of arbitrage opportunities is equivalent to the existence of a strictly positive vector of state-prices $\psi \in R_+^K$ such that $V = \Gamma\psi$.*

[...] Again, if the market is complete at $t = 0$, the K components of ψ solve a system of $K = N$ independent linear equations. In other words, the vector of state-prices is clearly uniquely defined as

$$\psi = \Gamma^{-1}V.$$

From Amaro de Matos §1.2.3 "Arbitrage Opportunities and Investment", p. 23 (PDF 41):

In fact, under the absence of arbitrage opportunities, the required investment $I^\top\theta$ in a portfolio θ must equal its value $V^\top\theta$. Otherwise an arbitrage opportunity exists, allowing an increase of current consumption by $(V-I)^\top\theta$, for fixed future consumption or, equivalently, increasing future consumption by $(V-I)^\top\theta(1+r)$, for fixed current consumption. In the language of the first section of this chapter, $(V-I)^\top\theta$ would be the net present value of such an arbitrage opportunity.

From Amaro de Matos §1.2.4 "Value and the Martingale Measure", p. 24 (PDF 42):

Assume the existence of an investment such that its payoff is a constant and therefore does not depend on state of nature to be realized in the future. Such an investment is said to be *risk-free*. As in section 1.1, assume that for each unit invested at time 0, a number $(1 + r)$ of consumption units are received at time 1. It then follows from Proposition 1.2.1 that

$$1 = (1 + r)\sum_{j=1}^{K} \psi_j$$

and the value vector may be rewritten as

$$V = \frac{1}{1+r}\Gamma\pi, \qquad (1.13)$$

where π is an K-dimensional vector with components $\pi_i = \psi_i / \sum_{j=1}^K \psi_j$.

What is asked. (a) The prices today of one unit of consumption in each of the two states, implied by the risk-free rate and project 1's price, assuming no arbitrage. (b) Whether an agent invests in project 1 (cost 5) and in project 2 (cost 2), given their payoffs.

Approach. Amaro de Matos' Proposition 1.2.1: no arbitrage $\Leftrightarrow$ a strictly positive state-price vector ψ with $V = \Gamma\psi$. With two states and two independent traded assets (the bond and project 1) the market is complete, so ψ is unique and prices any payoff. A project is worth undertaking when its market value exceeds its investment (positive NPV, §1.2.3).

Notation.

- States $s \in \{1, 2\}$ at $t = 1$; $\psi = (\psi_1, \psi_2)^\top$: state prices (value today of one unit paid in state s only).
- Risk-free asset (asset 0): price 1, payoff $1 + r = 1.01$ in each state, $r = 0.01$.
- Project 1: market value $V_1 = 6.7$, payoffs $(\Gamma_{11}, \Gamma_{12}) = (10, 5)$, investment $I_1 = 5$.
- Project 2: payoffs $(\Gamma_{21}, \Gamma_{22}) = (-2, 4)$, investment $I_2 = 2$, value V_2 to be found.
- Γ: payoff matrix (rows = assets, columns = states); V: vector of asset values; $q_s = (1 + r)\psi_s$: risk-neutral probabilities.

Step 1 — The market is complete. The traded assets are the risk-free asset and project 1:

$$\Gamma = \begin{pmatrix} 1.01 & 1.01 \\ 10 & 5 \end{pmatrix}, \qquad V = \begin{pmatrix} 1 \\ 6.7 \end{pmatrix}. \qquad (14.1)$$

Its determinant is $(1.01)(5) - (1.01)(10) = 5.05 - 10.10 = -5.05 \neq 0$, so the two payoff vectors are linearly independent and span both states. The number of independent assets equals the number of states, so the market is complete (Amaro de Matos §1.2.1) and the state-price vector, if it exists, is unique: $\psi = \Gamma^{-1}V$ (§1.2.2).

Step 2 — No arbitrage gives the pricing equations. By Proposition 1.2.1, absence of arbitrage is equivalent to a strictly positive ψ with $V = \Gamma\psi$. Row by row:

$$1 = 1.01\,\psi_1 + 1.01\,\psi_2, \tag{14.2}$$
$$6.7 = 10\,\psi_1 + 5\,\psi_2. \tag{14.3}$$

Step 3 — (a) Solve for ψ. Dividing (14.2) by 1.01:

$$\psi_1 + \psi_2 = \frac{1}{1.01} = 0.990099. \tag{14.4}$$

So $\psi_2 = 0.990099 - \psi_1$. Substituting into (14.3) and collecting the ψ_1 terms:

$$\begin{aligned} 6.7 &= 10\psi_1 + 5(0.990099 - \psi_1) \\ &= 10\psi_1 + 4.950495 - 5\psi_1 \\ &= 5\psi_1 + 4.950495, \\ 5\psi_1 &= 6.7 - 4.950495 = 1.749505, \\ \psi_1 &= \frac{1.749505}{5} = 0.349901. \end{aligned} \tag{14.5}$$

Substituting (14.5) into (14.4):

$$\begin{aligned} \psi_2 &= 0.990099 - (0.349901) \\ &= 0.640198. \end{aligned} \tag{14.6}$$

Both components are strictly positive, so by Proposition 1.2.1 (and exercise #1 of the same section) the prices V admit no arbitrage, and $\psi = (0.349901,\ 0.640198)$ is the unique state-price vector. Check: $10(0.349901) + 5(0.640198) = 3.49901 + 3.20099 = 6.70000$. Equivalently, the risk-neutral probabilities are $q_1 = (1.01)(0.349901) = 0.3534$ and $q_2 = (1.01)(0.640198) = 0.6466$.

Answer (a). $\psi = (\psi_1, \psi_2) \approx (0.3499,\ 0.6402)$.

Step 4 — (b) Value of project 2. In a complete market every payoff is priced by ψ:

$$\begin{aligned} V_2 &= \Gamma_{21}\psi_1 + \Gamma_{22}\psi_2 \\ &= (-2)(0.349901) + (4)(0.640198) \\ &= -0.699802 + 2.560792 \\ &= 1.860990. \end{aligned} \tag{14.7}$$

Justification by replication: a portfolio of a units of project 1 and b units of the bond pays project 2's payoffs if $10a + 1.01b = -2$ and $5a + 1.01b = 4$. Subtracting, $5a = -6$, so $a = -1.2$, and then $b = (4 - 5(-1.2))/1.01 = 10/1.01 = 9.900990$. Its cost is $(-1.2)(6.7) + (9.900990)(1) = -8.04 + 9.900990 = 1.860990$, equal to (14.7). If project 2's payoff traded at any other price, buying the cheaper of the two and selling the dearer would be an arbitrage, which Step 2 rules out.

Step 5 — NPVs. The NPV of a project is the value it creates today, $V_j - I_j$ (§1.2.3):

$$\mathrm{NPV}_1 = (6.7) - (5) = 1.7 > 0, \tag{14.8}$$
$$\mathrm{NPV}_2 = (1.860990) - (2) = -0.139010 < 0. \tag{14.9}$$

Step 6 — Every agent invests in project 1 and none in project 2. Take any agent with utility increasing in consumption. (i) Project 2: whatever future consumption the agent obtains by investing 2 in project 2 can be obtained by buying the replicating portfolio of Step 4 for 1.860990. The second route leaves $2 - 1.860990 = 0.139010$ more for consumption today with the same future consumption, so by monotonicity it is strictly preferred; no agent invests in project 2. (ii) Project 1:

investing 5 and selling claims to its payoff at the market value 6.7 raises today's consumption by 1.7 and leaves future consumption unchanged, which every agent strictly prefers. Hence agents invest in project 1 only, and this does not depend on their preferences.

Answer (b). No: $V_2 = 1.861 < 2$ ($\text{NPV}_2 = -0.139$), while $\text{NPV}_1 = 1.7 > 0$; agents invest in project 1 only.

Intuition. With two states and two independent traded assets, every state-contingent payoff has one market price. A project is worth doing if it produces its payoff more cheaply than the market sells it: project 1 does (5 against 6.7), project 2 does not (2 against 1.86).

Common mistakes.

- Forgetting the risk-free asset as the second pricing equation, leaving one equation for two unknowns.
- Using the bond payoff 1 instead of 1.01 (price 1 buys $1 + r$), which gives $\psi_1 + \psi_2 = 1$.
- Judging project 2 by its expected payoff or by the sign of Γ_{21} instead of comparing its state-price value with the investment of 2.

Book answer. Matches the book's answer (p. 247): $\psi \approx (0.35,\ 0.64)$, $V_2 = 1.86$, $\text{NPV}_1 = 1.7$, $\text{NPV}_2 = -0.14$, invest in project 1 only.

Ref: Amaro de Matos, §1.2.1 (one-period model, completeness), pp. 18–21; §1.2.2 (Proposition 1.2.1), pp. 22–23; §1.2.3 (arbitrage opportunities and investment), pp. 23–24; §1.2.4 (risk-neutral measure), p. 24.

15. Investment independent of preferences under uncertainty — Amaro de Matos 1.2.6 #4, p. 28

Statement.

Source: Amaro de Matos, Ch. 1, Exercise 1.2.6 #4, p. 28 (PDF 46) · Answer in book: yes, p. 245

Consider two agents, A and B, who live in a one-period economy defined by two points in time. They are given endowments at both points in time. Both agents have the possibility of making productive investments. However, there is uncertainty with respect to the return of this investment in the future. With probability p it is $F_1(\epsilon)$ and with probability $1 - p$ it is $F_2(\epsilon)$. As a consequence, an agent who makes a productive investment cannot be certain about the future consumption. With probability p agent i can consume y_i^1 and with probability $1 - p$ can consume y_i^2. Their preferences are described by a utility function $V_i = EU_i(x, y)$ where $U_i(x, y)$ has the properties described in section 1.1. Find sufficient conditions for the level of investment to be independent of the preferences and of the endowments of the agents.

Context needed to solve it *From Amaro de Matos §1.1.1 "The Robinson Crusoe Economy", pp. 5–6 (PDF 23–24) — the properties of $U(x, y)$:*

He lives in a one-period economy, defined by two points in time: now and the future (say, one year from now). [...] He is given endowments in a nonperishable good. Let X denote the initial endowment and Y denote the promised future endowment. Given this situation, Robinson faces a problem—namely, to choose how much to consume today, x, as compared to how much to consume in the second period, y. [...] Let us assume that his time preferences for consumption are fully characterized by a utility function $U(x, y)$ defined in R^2_+ [...]

Assumption 1.1.1. *The utility function $U(x, y)$ is continuous and twice differentiable in its two arguments.*

Within this context the derivatives with respect to the first and second arguments are denoted, respectively, by U_1 and U_2. The second derivatives are denoted accordingly, and we incorporate the usual assumption that

Assumption 1.1.2. *Utility functions increase in both arguments at decreasing rates.*

In other words, both U_1 and U_2 are positive, and both U_{11} and U_{22} are negative. [...]

Assumption 1.1.3. *The utility function $U(x, Y + X - x)$ is strictly concave as a function of x.*

From Amaro de Matos §1.1.3 "Production Opportunities", p. 8 (PDF 26):

Assumption 1.1.4. *At an investment level I, assume that an additional marginal investment has a marginal nonincreasing rate of return $\varphi(I)$, which is always positive.*

From Amaro de Matos §1.1.5 "Consumption and Investment with Capital Markets", pp. 11–12 (PDF 29–30) — the certainty benchmark (Fisher separation):

Assumption 1.1.5. *The infimum of the marginal decreasing rate of return $\varphi(I)$, which is always positive, is smaller than the interest rate r.*

[...] Given the endowment pair (X, Y), the existence of financial markets allows us to borrow money for investment. Let ϵ be the amount to be invested in the investment schedule. Thus, consuming x at the initial time $t = 0$, the amount available for the investment is $X - x$. If the required investment ϵ is greater than $X - x$, the amount $\epsilon - (X - x)$ must be borrowed from the bank. This implies that at time $t = 1$ the investor must pay back $[\epsilon - (X - x)](1 + r)$ to the bank. If, on the other hand, the required investment ϵ is smaller than $X - x$, the amount $(X - x) - \epsilon$ must be lent to the bank. Hence, in any case, the future consumption will be given by

$$y(x) = Y + \int_0^{\epsilon} [1 + \varphi(I)]\, dI - [\epsilon - (X - x)](1 + r) \qquad (1.5)$$

$$= X + Y - x + \int_0^{\epsilon} \varphi(I)\, dI - [\epsilon - (X - x)]\, r. \qquad (1.6)$$

Then the problem is to maximize the utility $U(x, y)$ subject to the possibilities allowed by the production schedule or, equivalently, maximizing $U(x, y(x))$ with $y(x)$ given by equation (1.5), subject to $\epsilon \geq 0$. Maximization must be performed with respect to x and ϵ. It is simple to check that the first-order condition with respect to ϵ does not depend on x and y, leading to an optimal value ϵ^* of ϵ given by

$$\epsilon^* = \begin{cases} 0 & r \geq \varphi(0) \\ \varphi^{-1}(r) & \varphi(0) \geq r. \end{cases}$$

This result may be interpreted as follows: The optimal level of investment ϵ^* does not depend on the preferences of the agents over the levels of consumption in both periods. [...]

Proposition 1.1.1 (Fisher separation). *The optimal production decision is determined by an objective market criterion independently of individuals' subjective preferences that define their consumption decisions.*

From Amaro de Matos §1.2.1 "One-Period Model", pp. 18–19 and 21 (PDF 36–37, 39) — expected utility and complete markets:

Uncertainty is thus described by a set of states belonging to a finite sample space $\Omega = \{w_1, w_2, \ldots, w_K\}$, where each state w_j occurs with probability $p(w_j)$, denoted for simplicity as p_j. [...] A commonly accepted utility representation is the *von Neumann-Morgenstern utility function*,

$$U^*(x, y) = EU(x, y) = \sum_{j=1}^{K} U(x, y_j)\, p_j,$$

also known as the *expected utility function*. [...] In other words, completeness of the market is equivalent to having the number of projects with linearly independent payoffs N equal to the number of states K.

What is asked. Two agents with different utilities and endowments can invest in a technology whose output is uncertain (F_1 or F_2). Find conditions under which both choose the same investment, i.e. the investment decision does not depend on who makes it: Fisher separation extended to uncertainty.

Approach. Under certainty, Fisher separation (Proposition 1.1.1) holds because a capital market lets each agent separate the production choice (maximise market value) from the consumption choice. Under uncertainty, the same works if agents can trade every state-contingent payoff at common prices: complete markets with state prices ψ (Proposition 1.2.1). Then each agent's wealth is endowment value plus the market value created by investing, and every agent maximises the latter. We prove sufficiency directly, then show the book's first-order-condition route.

Notation.

- Agents $i \in \{A, B\}$; dates $t = 0, 1$; states $s \in \{1, 2\}$ at $t = 1$ with probabilities $p_1 = p$, $p_2 = 1 - p$, $0 < p < 1$.
- $X_i > 0$: endowment at $t = 0$; $Y_i^s \geq 0$: endowment at $t = 1$ in state s (it may differ across states).
- $\epsilon_i \geq 0$: productive investment at $t = 0$; $F_s(\epsilon)$: its output at $t = 1$ in state s (same technology for both agents).
- x_i: consumption at $t = 0$; y_i^s: consumption at $t = 1$ in state s.

- $U_i(x,y)$: utility with the properties of §1.1 (continuous, twice differentiable, $U_x, U_y > 0$, concave); $V_i = p\,U_i(x,y^1) + (1-p)\,U_i(x,y^2)$.
- r: risk-free rate; $\psi_s > 0$: state prices; z_i^s: holding of the Arrow security paying 1 in state s only (any sign).
- $\Pi(\epsilon) \equiv \psi_1 F_1(\epsilon) + \psi_2 F_2(\epsilon) - \epsilon$: market value created by investing ϵ.

Claim. The following conditions are sufficient for both agents to choose the same investment ϵ^*, independent of U_A, U_B and of the endowments:

(S1) *Complete, frictionless markets at $t=0$:* besides the risk-free bond there is a traded security whose payoff is not proportional to $(1,1)$, so that (with the bond) any state-contingent payoff can be bought; no short-sale limits; agents take prices as given. Equivalently, the two Arrow securities trade at prices ψ_1, ψ_2.

(S2) *No arbitrage:* $\psi_1, \psi_2 > 0$ (Proposition 1.2.1); the bond is priced consistently, $\psi_1 + \psi_2 = 1/(1+r)$.

(S3) *Technology:* F_1, F_2 are differentiable, nondecreasing and concave, at least one strictly concave, and there is $\bar\epsilon$ with $\psi_1 F_1'(\bar\epsilon) + \psi_2 F_2'(\bar\epsilon) < 1$ (the analogue of Assumption 1.1.5: beyond some scale, investing returns less than the market).

Then ϵ^* is the unique maximiser of Π on $[0,\infty)$, characterised by

$$\psi_1 F_1'(\epsilon^*) + \psi_2 F_2'(\epsilon^*) = 1 \quad \text{if } \psi_1 F_1'(0) + \psi_2 F_2'(0) > 1, \qquad \epsilon^* = 0 \text{ otherwise.} \tag{15.1}$$

Step 1 — Budget constraints. Agent i consumes, invests and buys Arrow securities at $t=0$, and consumes endowment, output and security payoffs at $t=1$:

$$x_i + \epsilon_i + \psi_1 z_i^1 + \psi_2 z_i^2 = X_i, \tag{15.2}$$

$$y_i^s = Y_i^s + F_s(\epsilon_i) + z_i^s, \qquad s = 1,2. \tag{15.3}$$

Lending b at the risk-free rate is the case $z_i^1 = z_i^2 = (1+r)b$, which by (S2) costs $(\psi_1+\psi_2)(1+r)b = \frac{1}{1+r}(1+r)b = b$; so the bond is included in (15.2)–(15.3).

Step 2 — Collapse to one wealth constraint. From (15.3), $z_i^s = y_i^s - Y_i^s - F_s(\epsilon_i)$. Substituting into (15.2):

$$x_i + \epsilon_i + \psi_1\big(y_i^1 - Y_i^1 - F_1(\epsilon_i)\big) + \psi_2\big(y_i^2 - Y_i^2 - F_2(\epsilon_i)\big) = X_i.$$

Moving the terms that do not involve consumption to the right:

$$\begin{aligned} x_i + \psi_1 y_i^1 + \psi_2 y_i^2 &= X_i + \psi_1 Y_i^1 + \psi_2 Y_i^2 + \psi_1 F_1(\epsilon_i) + \psi_2 F_2(\epsilon_i) - \epsilon_i \\ &= \underbrace{X_i + \psi_1 Y_i^1 + \psi_2 Y_i^2}_{\text{endowment wealth } W_i^0} + \Pi(\epsilon_i). \end{aligned} \tag{15.4}$$

Because z_i^1, z_i^2 are unrestricted (S1), for a given ϵ_i every consumption plan satisfying (15.4) can be reached (choose z_i^s from (15.3)), and every reachable plan satisfies (15.4). So the agent's choice set, given ϵ_i, is the budget line (15.4) with wealth $W_i^0 + \Pi(\epsilon_i)$.

Step 3 — Any optimal investment maximises Π. Let ϵ^* maximise Π (it exists by Step 4) and suppose an agent's optimum used some ϵ' with $\Pi(\epsilon') < \Pi(\epsilon^*)$, with consumption (x', y'^1, y'^2) satisfying (15.4) at ϵ'. Define

$$x'' = x' + \big(\Pi(\epsilon^*) - \Pi(\epsilon')\big) > x'. \tag{15.5}$$

Substituting (x'', y'^1, y'^2) and ϵ^* into the left side of (15.4):

$$\begin{aligned} x'' + \psi_1 y'^1 + \psi_2 y'^2 &= \big(x' + \Pi(\epsilon^*) - \Pi(\epsilon')\big) + \psi_1 y'^1 + \psi_2 y'^2 \\ &= \big(x' + \psi_1 y'^1 + \psi_2 y'^2\big) + \Pi(\epsilon^*) - \Pi(\epsilon') \\ &= \big(W_i^0 + \Pi(\epsilon')\big) + \Pi(\epsilon^*) - \Pi(\epsilon') \\ &= W_i^0 + \Pi(\epsilon^*), \end{aligned}$$

so the plan is feasible with ϵ^*. Since $U_x > 0$ (Assumption 1.1.2), $V_i(x'', y'^1, y'^2) > V_i(x', y'^1, y'^2)$, contradicting optimality. Hence every agent's optimal investment lies in $\arg\max_{\epsilon \geq 0} \Pi(\epsilon)$. The argument used only that utility increases in x: not the form of U_i, not (X_i, Y_i^1, Y_i^2), and not even p.

Step 4 — The maximiser is unique and solves (15.1)**.** Differentiating Π term by term (ψ_s are constants):

$$\Pi'(\epsilon) = \psi_1 F_1'(\epsilon) + \psi_2 F_2'(\epsilon) - 1, \tag{15.6}$$

$$\Pi''(\epsilon) = \psi_1 F_1''(\epsilon) + \psi_2 F_2''(\epsilon) < 0, \tag{15.7}$$

where (15.7) uses $\psi_s > 0$ (S2) and concavity with one F_s strictly concave (S3). So Π is strictly concave and has at most one maximiser. By (S3), $\Pi'(\bar{\epsilon}) < 0$, and Π' is decreasing, so Π falls beyond $\bar{\epsilon}$: a maximiser exists on the compact set $[0, \bar{\epsilon}]$. The Kuhn–Tucker conditions for $\max_{\epsilon \geq 0} \Pi$ are $\Pi'(\epsilon^*) \leq 0$, $\epsilon^* \geq 0$, $\epsilon^* \Pi'(\epsilon^*) = 0$. If $\Pi'(0) > 0$, then $\epsilon^* = 0$ violates $\Pi'(\epsilon^*) \leq 0$, so $\epsilon^* > 0$ and $\Pi'(\epsilon^*) = 0$, which with (15.6) is the first case of (15.1); if $\Pi'(0) \leq 0$, strict concavity gives $\Pi'(\epsilon) < 0$ for all $\epsilon > 0$, so $\epsilon^* = 0$. Both agents therefore choose $\epsilon_A = \epsilon_B = \epsilon^*$. □

Step 5 — Link with the agents' first-order conditions (the book's route). Substituting (15.2)–(15.3) into V_i, i.e. $x_i = X_i - \epsilon_i - \psi_1 z_i^1 - \psi_2 z_i^2$ and $y_i^s = Y_i^s + F_s(\epsilon_i) + z_i^s$, and differentiating with respect to z_i^s and ϵ_i at an interior optimum (chain rule: $\partial x_i / \partial z_i^s = -\psi_s$, $\partial y_i^s / \partial z_i^s = 1$, $\partial x_i / \partial \epsilon_i = -1$, $\partial y_i^s / \partial \epsilon_i = F_s'$):

$$\frac{\partial V_i}{\partial z_i^s} = -\psi_s \bar{U}_x + p_s U_y^s = 0 \implies \psi_s = \frac{p_s U_y^s}{\bar{U}_x}, \tag{15.8}$$

$$\frac{\partial V_i}{\partial \epsilon_i} = -\bar{U}_x + p U_y^1 F_1'(\epsilon_i) + (1-p) U_y^2 F_2'(\epsilon_i) = 0, \tag{15.9}$$

where $\bar{U}_x \equiv p\, \partial_x U_i(x_i, y_i^1) + (1-p)\, \partial_x U_i(x_i, y_i^2) > 0$ and $U_y^s \equiv \partial_y U_i(x_i, y_i^s) > 0$. Dividing (15.9) by $\bar{U}_x$ and substituting (15.8):

$$-1 + \Big(\frac{p U_y^1}{\bar{U}_x}\Big) F_1'(\epsilon_i) + \Big(\frac{(1-p) U_y^2}{\bar{U}_x}\Big) F_2'(\epsilon_i) = 0,$$

$$(\psi_1) F_1'(\epsilon_i) + (\psi_2) F_2'(\epsilon_i) = 1, \tag{15.10}$$

which is (15.1). Each agent's marginal rates of substitution are individual, but in a complete market each agent trades until they equal the *common* prices ψ_s, and only those prices enter (15.10). With $q_s = (1+r)\psi_s$ (so $q_1 + q_2 = 1$ by (S2)), multiplying (15.10) by $1+r$ gives $q_1 F_1'(\epsilon^*) + q_2 F_2'(\epsilon^*) = 1 + r$: the expected marginal return under the risk-neutral measure equals the gross risk-free rate, the uncertainty version of $\phi(\epsilon^*) = r$ in §1.1.5.

Step 6 — Why completeness is needed, and a second sufficient condition. With only the bond, $z_i^1 = z_i^2$ is forced and (15.8) is replaced by the single condition $\bar{U}_x = (1+r)\big(pU_y^1 + (1-p)U_y^2\big)$. Dividing (15.9) by it gives

$$m_i F_1'(\epsilon_i) + (1 - m_i) F_2'(\epsilon_i) = 1 + r, \qquad m_i \equiv \frac{p U_y^1}{p U_y^1 + (1-p) U_y^2}, \tag{15.11}$$

and m_i depends on agent i's utility and, through y_i^s, on endowments: investment is no longer separated from preferences. The exception is when the marginal return carries no uncertainty, $F_1'(\epsilon) = F_2'(\epsilon) \equiv F'(\epsilon)$ for all ϵ (the book's "trivial uncertainty"). Substituting into (15.11):

$$m_i \left(F'(\epsilon_i)\right) + (1 - m_i) \left(F'(\epsilon_i)\right) = 1 + r,$$

$$(m_i + 1 - m_i) F'(\epsilon_i) = 1 + r,$$

$$F'(\epsilon_i) = 1 + r,$$

the same for every agent. So either (S1)–(S3), or a bond market plus nonstochastic marginal returns (with concave F), is sufficient.

Answer (conditions). Complete, frictionless markets with strictly positive state prices (no arbitrage) and a concave technology: then both agents choose ϵ^* solving $\psi_1 F_1'(\epsilon^*) + \psi_2 F_2'(\epsilon^*) = 1$, i.e. $q_1 F_1' + q_2 F_2' = 1 + r$, independent of U_i and endowments. (Alternative: bond market and $F_1' = F_2'$.)

Intuition. Fisher separation survives uncertainty when markets let every agent reshape the project's risky output into whatever consumption pattern they like at common prices. Then the best project is the one with the highest market value, and all agents, however patient or risk averse, agree on it. Without complete markets each agent values the risky output with private marginal utilities, and investment depends on who is investing.

Common mistakes.

- Weighting F_1' and F_2' by the true probabilities $p, 1-p$ instead of the state prices (or risk-neutral probabilities): that is only right for risk-neutral agents.
- Thinking a risk-free bond alone is enough: with a bond only, each agent's weights m_i in (15.11) depend on their own marginal utilities.
- Stating the first-order condition without the conditions (concavity, a scale bound) that make the optimum exist and be unique.

Book answer. Matches the book's answer (pp. 245–247): a unique state-price vector (two states and two independent assets) gives an investment rule, its (C.3), $q(\theta_1)F_\epsilon(\epsilon, \theta_1) + q(\theta_2)F_\epsilon(\epsilon, \theta_2) = 1 + r$, that involves neither U, x, y nor the endowments. The book identifies its weights with $p_s U_y^s$; they equal (15.10)'s $q_s = (1 + r)\psi_s$ once normalised to sum to one. The solution above adds the concavity and scale conditions (S3) that make ϵ^* exist and be unique, and proves the result without differentiating U (Step 3).

Ref: Amaro de Matos, §1.1.1 (Assumptions 1.1.1–1.1.3), pp. 5–6; §1.1.5 (Proposition 1.1.1, Fisher separation), pp. 11–13; §1.2.1–1.2.3 (state prices, completeness, Proposition 1.2.1, arbitrage and investment), pp. 18–24.